

Independent Power Projects in Sub-Saharan Africa: Investment trends and policy lessons



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ABSTRACT

Sub-Saharan Africa is in urgent need of more power. Private sector investment is key to achieving this. Along with Chinese-funded projects, Independent Power Projects (IPP) represent the fastest growing sources of power investment in Sub-Saharan Africa. IPP investment flows show little concern for electricity market structures, but are more likely to gravitate to countries with strong planning, procurement and contracting capacity, as well as good regulatory quality. Data from the continent also shows a variety of ownership and financing structures for IPPs, but generally development financing institutions (DFIs) play an important part in mitigating risk and bringing in private financiers. We also see renewable energy breaking through on the continent - both in scale and price. This breakthrough is in part being facilitated by competitive procurement or auctions, which deliver lower prices and increased transparency when compared with renewable energy feed-in tariffs or directly negotiated contracts. These developments have important policy implications, highlighting the need for: dynamic, least-cost planning, linked to the timely initiation of the competitive procurement of new generation capacity; the building of effective regulatory capacity; and appropriate risk mitigation mechanisms. Such efforts promise to promote sustainable economic and social development across the continent.

1. The need for Independent Power Projects in Africa

Sub-Saharan Africa (SSA) has a severe shortage of power. In 2014 the 49 Sub-Saharan African countries, with a combined population of more than 800 million, had less generating capacity (92 GW) installed than Spain (106 GW), a country with a population of 45 million (Findt et al., 2014; U.S. EIA, 2014). What further sharpens this contrast is the fact that more than half of the region's installed capacity is based in a single country: South Africa. The remaining 46 GW is therefore shared among the remaining 48 countries in the region, with only 14 countries having power systems larger than 1 GW. Put another way: installed capacity in Sub-Saharan Africa is 44 MW per million people, compared with 192 MW per million people in India, 590 MW in Latin America, and 815 MW in China (U.S. EIA, 2014). Electricity demand is set to double by 2030, and triple by 2040 (International Energy Agency, 2015). A recent report by McKinsey estimates that more than \$490 billion will need to be invested in additional power generation capacity by 2040 to meet projected demand (Castellano et al., 2015). Approximately \$45.6 billion was invested in electric power generation in Sub-Saharan Africa between 1990 and 2013; excluding South Africa,

this total drops to \$31.3 billion (Eberhard et al., 2016). Existing investment levels are therefore far below what is required, calling for increased private sector involvement (Eberhard and Shkaratan, 2012). Independent Power Projects, or IPPs, are the main source of private investment in the African power sector (Eberhard and Gratwick, 2013). While these entities are having a significant impact on the African power sector landscape, relatively little is known about their related outcomes and the factors driving and underpinning these infrastructure investments.

The research questions that we aim to address, are:

- What are the main power sector & IPP investment trends in Sub-Saharan Africa?
- Why are some countries more successful in attracting private power investments than others?
- What are the different IPP types (ownership structures, technology choices, procurement methods) in Sub-Saharan Africa, and what are the related outcomes?
- What are the key lessons for scaling up investment in power generation in Africa?

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2. Methods

We define IPPs as power projects that are, in the main, privately developed, constructed, operated and owned; have a significant proportion of private finance; and have long-term power purchase agreements with a utility or another off-taker. IPPs included in this study are all greenfield, grid-connected installations of 5 MW (MW) or greater that have reached financial close, are under construction, or are in operation. A significant amount of data on power projects has been collected and analyzed for this study. Sources include a series of World Bank databases, including the Private Participation in Infrastructure (PPI) database; data from the Energy Information Administration (EIA); and databases prepared by Aid Data and the OECD, among others. In addition, the authors have conducted primary and secondary source research, particularly on individual Independent Power Projects.

Apart from the above-noted data sources, the analysis of IPP types and outcomes, as well as the identification of lessons learned, is based primarily on original, in-depth case studies carried out in five countries, namely Kenya, Nigeria, South Africa, Tanzania, and Uganda. The five case study countries were selected because they present the largest and most diversified experience with IPPs over the longest time period, accounting for around 80% of IPP investment in Sub-Saharan Africa. Each country has developed four or more IPPs, a fact that facilitates an assessment of enabling policies and regulatory frameworks, planning and procurement practices, and lessons learned. All five countries have been host to IPPs with different technology bases, which allows for a relatively in-depth evaluation of cost and reliability. Finally, each country has a mix of directly negotiated and competitively bid projects, which has the potential to shed light on which procurement methods are more effective.

2.1. Data limitations

Although an unprecedented body of data and case histories have been collected and analyzed, data limitations remain. Information concerning the composition of investments by funding source; the terms of IPP contracts (which remain mostly confidential); and the size, composition, and types of investment from emerging financiers (notably China) was gathered from various sources and triangulated. For Chinese data specifically, the authors used Aid Data as a starting point. Additional secondary source research was conducted, and then actual projects were verified with stakeholders in each of the study countries. However, because nearly every Chinese-funded generation project is directly negotiated with the government of a given African country, limited public data is available.

Due to a lack of available data, government and utility megawatts and investments have largely been derived by (i) subtracting the megawatt totals of IPPs, Chinese, official development assistance (ODA), and multilateral finance institutions, and development finance institutions, and then (ii) using the Energy Information Administration's corresponding data on "megawatts installed by technology" per country to determine residual megawatts per technology (U.S. EIA, 2014), and finally (iii) ascribing an investment value, based on average costs per technology in Sub-Saharan Africa. Wherever possible, efforts have been made to verify the megawatts and the technology with known projects undertaken by the government.

The focus of this paper is on power generation, as opposed to the transmission and distribution (T&D) of electricity. While inadequate T&D is clearly a constraint on any effort to widen service access, countries must have sufficient generation capacity to be able to serve new customers, improve welfare, and accelerate economic development. Also, a detailed discussion of the environmental externalities attached to specific IPP technologies—which pose growing concerns outside the purview of this paper.

Finally, South Africa's size and prominence in the generation of

Sub-Saharan Africa's electric power is noteworthy and hence efforts have been made to present Sub-Saharan African tallies with and without South Africa.

3. Trends in power generation investment in Sub-Saharan Africa

3.1. Investment trends

Power investments in Sub-Saharan Africa between 1990 and 2013 were far below requirements: only 15.63 GW net was added across the region, excluding South Africa (U.S. EIA, 2014). The 1990s saw a mere 1.84 GW of new capacity installed. Investment picked up since 2000, with an additional 13.8 GW installed in the region. Around 94% of this capacity has been added in only 15 countries, with the rest adding hardly any capacity at all, and some even losing capacity as a result of civil wars or poor maintenance (U.S. EIA, 2014).

While historically public utilities have been the major sources of new investment, this trend is changing. Most African governments are unable to fully fund their power needs, and most utilities do not have investment-grade ratings and so cannot raise sufficient debt at affordable rates (Eberhard and Gratwick, 2013). Official Development Assistance (ODA) and development finance institutions (DFIs) have only partially filled the funding gap. The fastest growing sources of finance for Africa's power sector are now private investments in IPPs and Chinese funding (Eberhard et al., 2016). Nevertheless, around 50% of investment in the African power sector is still coming from the public sector, but it has remained stagnant over the period analyzed. In addition, concessional DFI funding, ODA and Arab funding represent a small portion of the overall funding picture, with no real growth. The continent therefore seems set to increase its dependence on private and foreign (Chinese) investments to fund its power generation needs in the near and medium term (Fig. 1).

3.2. Chinese funding

While not the explicit focus of this paper, the growing size and prominence of China's involvement in the African power sector warrants some discussion. Chinese funded generation assets represent an important area of significant capacity additions in Sub-Saharan Africa, totalling 34 projects in 19 countries between 1990 and 2014 (Fig. 2). Taken together, these represent a total of 7.5 GW in installed capacity, with most capacity added in the years 2009–2014 (Eberhard et al., 2016). According to the International Energy Agency, Chinese capacity additions account for more than 30% of new capacity

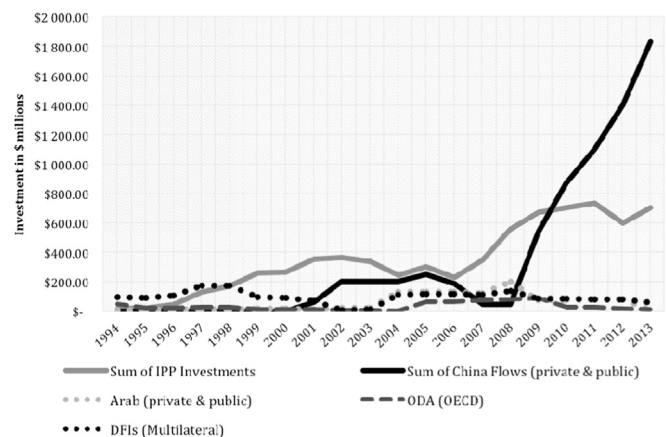


Fig. 1. Investments in Power Generation, Five-Year Moving Average: Sub-Saharan Africa (Excluding South Africa), 1994–2013. Note: DFI = Development Finance Institutions; IPP = Independent Power Project; ODA = Official Development Assistance; OECD = Organization for Economic Co-operation and Development.

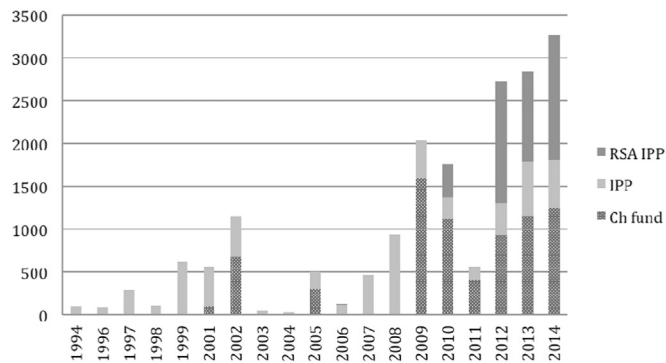


Fig. 2. Comparison of Chinese-funded power projects and IPPs, by generation capacity: Sub-Saharan Africa, 1994 – 2014. Note: No IPPs recorded for 1995 or 2000, which explains the absence of those years in the figure. IPP = Independent Power Project; RSA = Republic of South Africa.

Source: compiled by the authors, based on various primary and secondary sources.

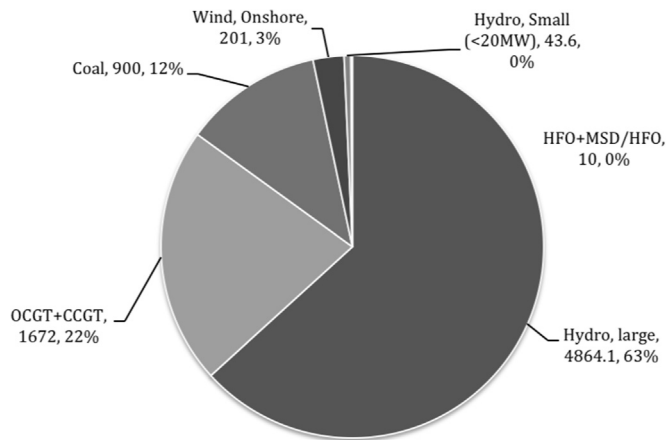


Fig. 3. Chinese-supported power project capacity (% of MW), by technology: Sub-Saharan Africa, 2001 – 2014. Note: CCGT = combined-cycle gas turbine; HFO = heavy fuel oil; MSD = medium-speed diesel; MW = megawatt; OCGT = open-cycle gas turbine. Source: Compiled by the authors, based on various primary and secondary sources.

additions in Sub-Saharan Africa between 2010 and 2015 (International Energy Agency, 2016). The agency also reports that Chinese contractors have built or are set to build more than 17 GW of new capacity in the 2010 – 2020 period (International Energy Agency, 2016). It is however important to note that there is a considerable and growing gap between Chinese funding for African infrastructure and the involvement of Chinese contractors: close to 50% of the continent's infrastructure contracts in 2014 are being awarded to Chinese companies, while Chinese funding accounts for less than 5% of the continent's overall infrastructure financing in the same period (Huang and Chen, 2016).

Chinese power sector investments in Sub-Saharan Africa do not appear to follow an expected pattern, with no clear correlation between the resource-wealth of a country and Chinese-backed investments.^a This is in line with findings by Brautigam (2009) as well as Dreher and Fuchs (2006), which challenge the notion of Chinese aid and investments in Africa being primarily motivated by resource-based considerations. What is however apparent is a preponderance for large (> 50 MW) hydro-power projects (Fig. 3) (Eberhard et al., 2016), compos-

^a Chinese-funded power generation projects exist in the following 19 countries: Botswana, Cameroon, the Central African Republic, the Democratic Republic of Congo, the Republic of Congo, Côte d'Ivoire, Equatorial Guinea, Ethiopia, Gabon, Ghana, Guinea, Liberia, Mali, Nigeria, Sudan, Togo, Uganda, Zambia and Zimbabwe.

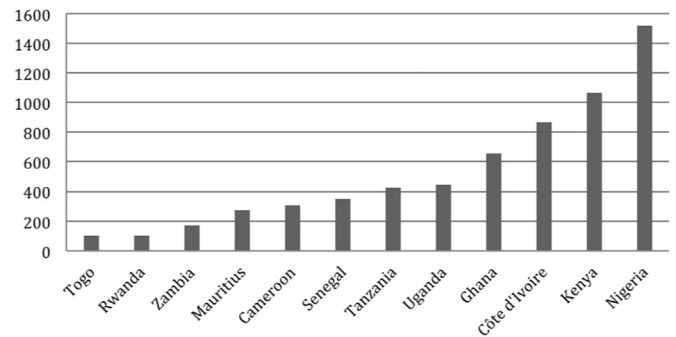


Fig. 4. Countries with the most independent power project capacity in Sub-Saharan Africa, excluding South Africa, 1994 – 2014 (MW).

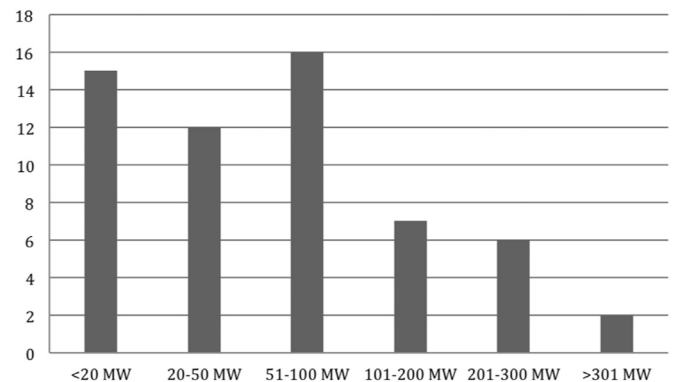


Fig. 5. Number of Independent Power Projects in various size categories (megawatt), to have reached financial close as of 2014.

Source: Compiled by authors, based on utility data, primary sources, and the PPI database.

ing 63% of Chinese-funded capacity between 2001 and 2014 and for which Chinese engineering, procurement and construction (EPC) contractors have become renowned worldwide. This is in line with findings from the IEA's investigation into Chinese capacity additions, which finds that 49% of all power projects by Chinese companies for the 2010 – 2020 period is hydropower focused (International Energy Agency, 2016).

3.3. IPPs in Africa

Africa's experience with IPPs started in 1994 in Côte d'Ivoire, followed by Kenya in 1996 and Mauritius in 1997 (Eberhard and Gratwick, 2011). IPPs have since spread to 18 countries (excluding South Africa), with 59 projects having been implemented. Most of these projects are concentrated in only a handful of countries (Eberhard et al., 2016) (Fig. 4). From 2011, investments began taking off, with the years since (2011–2014) constituting the largest and most sustained investment cycle to date, representing 14 projects (excluding South Africa), \$4.9 billion in investment and an additional 2.1 GW in capacity. In total, IPPs in Sub-Saharan Africa represent more than \$11.12 billion in investments and 6.8 GW of installed capacity. Adding South Africa's 92 renewable energy IPPs brings this total to 151 projects, totalling more than \$30 billion in investment and more than 12 GW in installed capacity. IPPs however still represent a minority of total generation capacity, mainly complementing state-owned utilities.

IPPs in Sub-Saharan Africa range in size from a few megawatts to around 600 MW – though the majority of projects, about two thirds, are smaller than 100 MW (Eberhard et al., 2016) (Fig. 5). The overwhelming capacity (82%) is thermal – mostly open and combined-cycle gas turbines. However, there is important growth in renewables –

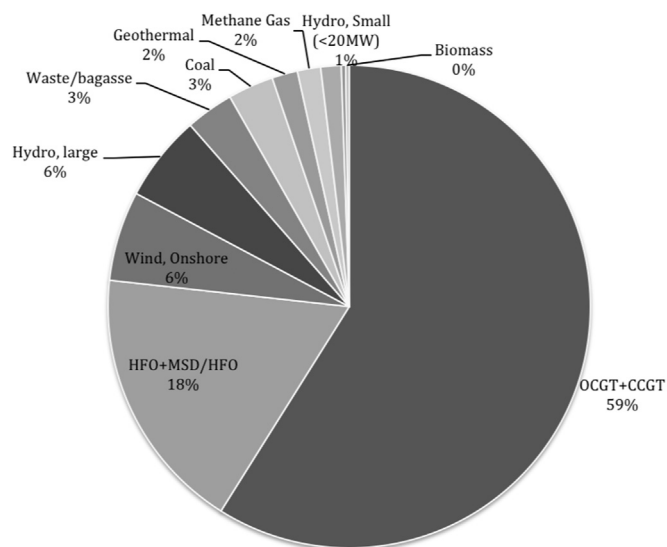


Fig. 6. Independent Power Project Capacity (% of MW), by Technology: Sub-Saharan Africa (Excluding South Africa), 1994 – 2014. Note: CCGT = combined-cycle gas turbine; HFO = heavy fuel oil; MSD = medium-speed diesel; MW = megawatts; OCGT = open-cycle gas turbine.

Source: Compiled by the authors, based on utility data, primary sources, and the Private Participation in Infrastructure (PPI) database.

most notably wind, solar PV and concentrated solar power (CSP) (Fig. 6).

There has been a wide variety of African IPP sponsors and debt providers, though a few have backed multiple projects. Table 1 highlights specific IPPs from the five country case studies (excluding South Africa, which is treated separately due to its sheer size).

While state institutions have invested in some IPPs—for example, the Nigerian National Petroleum Corporation (Okpai and Afam) and the government of Uganda (Bujagali), as well as the Kenya Power Staff Pension Fund (Iberafrica)—private sponsors are prominent. Private African partners are present in numerous projects and recently have even taken majority or full equity, as in the case of Aba Integrated (Nigeria), Gulf and Triumph (Kenya), and Tororo and Buseruka (Uganda). Following this, the most conspicuous equity sponsor, Globeleg, hails from Europe, and there are 15 other European entities, such as Aldwych and Wartsila, as well as numerous European bilateral DFIs, such as the Norwegian Investment Fund for Developing Countries (Norfund), the Netherlands Development Finance Company (FMO), and the Danish Investment Fund for Developing Countries (IFU). North America sponsors (primarily from the United States) are significantly fewer, at only seven, followed by South Asia (one), Southeast Asia (one), and the Middle East (one). Equity is also held by multilateral agencies, namely, the International Finance Corporation (IFC) and new infrastructure funds: for example, the African Infrastructure Investment Fund (AIIF) managed by a South African life insurer.

3.4. ODA and DFI financing

The sustained upward trend in IPP investments on the continent in the past 5 years is all the more impressive if one considers that of the 18 Sub-Saharan African countries with IPPs, only South Africa, Botswana and Mauritius have investment-grade ratings (Mecagni et al., 2014). This limits the possibilities of traditional project financed IPP deals. It is in this context where development finance institutions (DFIs) that invest in the private sector (e.g. International Finance Corporation (IFC), the Netherlands Development Finance Company (FMO), Proparco, the Norwegian Investment Fund for Developing Countries (Norfund)) have played a significant role in funding IPPs

(Fig. 7) – bringing to financial close projects that would otherwise not have been viable.^b

4. Factors that support IPPs and their success

Despite the increase in funding, it is still only a small number of countries that are attracting IPPs. Why have some countries been more successful than others in attracting private investment? The following are some of the emergent determining factors for IPP investment flows.

4.1. Power markets

Sub-Saharan African utilities are often not credit-worthy off-takers, struggling to keep the lights on and unable to increase capacity (Eberhard and Shkaratan, 2012). The aim of power sector reform in developing countries is primarily to improve utility performance – both technical & commercial – and in the process attract more investment, to provide sustainable, affordable, adequate and reliable services for all, and power economic growth (Gratwick and Eberhard, 2008).

The standard model of power sector reform – unbundle, privatise, create regulatory institutions, and create markets (Victor and Heller, 2007) – is partly built on the premise that privatization will ultimately attract greater sector investment (Jamasb et al., 2005). The unbundling of generation from Transmission and Distribution (T & D) was initially understood as a key reform element, and one that arguably should even precede the introduction of IPPs to ensure fairness in the contracting and dispatching of IPPs (versus the utility off-taker's own generation). However, nowhere in Africa is full wholesale or retail competition in place (Kapika and Eberhard, 2013). Instead, we find various hybrid models – where public and private investment co-exist – implemented across the continent. IPPs are found in very different market structures (Fig. 8), with no clear correlation between the level or sequencing of reform and IPP investment. The majority of IPPs are in fact found in countries with vertically integrated utilities. This does not negate the importance of sector reform, especially from sustainability, performance and governance points of view; it does however pose an important question regarding what was once assumed to be driving private power sector investment. If power sector restructuring is not essential for attracting investment, then what is?

4.2. Independent regulation

By definition, IPPs are investment transactions regulated by the underlying contracts. Regulations at the sector level are also important in defining the rules of the game and ultimately shaping the enabling environment for IPPs (Findt et al., 2014). The establishment of energy regulators has been the most widespread power sector reform element in Sub-Saharan Africa, with more than half of the region's countries having such an agency (Kapika and Eberhard, 2013). As a rule, the countries with the most IPPs all have an independent regulatory authority.

An independent regulator brings with it oversight capacity and could potentially enforce the competitive procurement of IPPs. This has unfortunately not been the case in many Sub-Saharan African countries, and we are increasingly seeing regulatory risk serve as a further disincentive to investment. Merely having a regulator is thus not sufficient; rather, it is the quality of regulation produced and enforced by this regulator that is critical for attracting private investment. Transparent, fair and accountable regulators that produce

^b As stated before, only three countries in Sub-Saharan Africa have investment-grade credit ratings: South Africa, Mauritius and Botswana. Of these three, only Mauritius and South Africa have IPPs. Because of South Africa's distorting effect, Fig. 7 excludes data from South Africa. Mauritius has six IPPs, but only one has any DFI involvement. This is the Belle Vue Power Plant, a 71.2MW coal/bagasse plant that reached financial close in 1998 and was provided with a US\$17 million loan from the EIB.

Table 1

Project sponsors and debt holders in selected case studies.

Source: Compiled by authors, based on various primary and secondary source data.

Project	Equity partners (country, % of equity held)	Procurement	Contract change Y=yes/N=no	Equity turnover (#) ^a
Kenya				
Westmont	Equity: Westmont (Malaysia, 100%); sought to sell plant since 2004; ultimately towed back to Malaysia Debt: equity financed	DN	Not extended	—
Iberafrica	Equity: Union Fenosa (Spain, 80%), KPLC Pension Fund (Kenya, 20%) since 1997 Debt: Union Fenosa (\$12.7 million in direct loans and guaranteed \$20 million); KPLC Staff Pension Fund (\$9.4 million in direct loans and guaranteed \$5 million through local Kenyan bank)	DN	Y	0
OrPower4	Equity: Ormat (USA, 100%) since 1998 Debt: Equity financed until 2009, European DFIs \$105 million loan in 2009, then OPIC loan of \$310 million drawn down in 2012–13	ICB	Y	0
Tsavo	Equity: Cinergy (USA) and IPS (Int'l) jointly owned 49.9%; Cinergy sold to Duke Energy (USA) in 2005, CDC/Globeleq (UK, 30%), Wartsila (Finland, 15%), and IFC (Int'l, 5%) retain remaining shares since 2000 Debt: IFC own account (\$16.5 million), IFC syndicated (\$23.5 million), CDC own account (\$13 million), DEG own account (€11 million), DEG syndicated (€2 million)	ICB	N	1
Rabai	Equity: Aldwych-International (Netherlands, 34.5%), BWSC (Danish, but owned by Mitsui of Japan, 25.5%), FMO (Netherlands, 20%), IFU (Danish bilateral lender, 20%) Debt: FMO (\$126 million), Proparco and EAIF (25% each), DEG (15%), European Financing Partners (10%)	ICB	N	0
Mumias	Equity: Mumias Sugar Company Limited (100%/Kenya) Debt: Not available	DN	N	0
Thika	Equity: Melec Powergen (part of Maletec grp) (90%/Lebanon) Debt: AfDB (€28 million), IFC (€28 million), ABSA Capital (€28 million)	ICB	N	0
Triumph	Equity: Broad Holding (Kenya), Interpel Investments (Kenya), Taceflex (Kenya), Southern Inter-trade (Kenya) Debt: Industrial and Commercial Bank of China (ICBC) (\$80 million), and Kenya's CFC Stanbic Bank (\$28 million) (of which Standard Bank is the parent, in which ICBC has a 20% stake)	ICB	N	0
Gulf	Equity: Consortium of local investors, namely Gulf Energy Ltd and Noora Power Ltd Debt: \$76 million in long-term debt financing (IFC A Loan, and commercial lending through IFC B Loan and OPEC Fund for International Development)	ICB	N	0
Kinangop	Equity: Aeolus Kenya, AIIF2, majority owner (South Africa/Mauritius), Norfund (Norway) Debt: Kenyan CFC Stanbic Project stalled.	REFIT	N	0
Turkana	Equity: KP & P Africa BV (Netherlands) with Aldwych International (Netherlands) Debt: (foreign and local): AfDB, EIB, the Standard Bank of South Africa, Nedbank, FMO, Proparco, East African Development Bank (EADB), PTA Bank, EKF, Triodos, and DEG. The project's debt raising for the generation project was led by the AfDB, as mandated lead arranger, with the Standard Bank of South Africa and Nedbank Limited as coarrangers	DN	N	0
Nigeria				
AES	Equity: Enron (USA, 100%) sold to AES (95%) and YFP (Nigeria, 5%) in 2000	DN	Y	1
Barge	Debt: \$120 million loan (foreign and local): RMB (South Africa), FMO, African Export Import Bank, Diamond Bank Nigeria, Fortis Bank, KfW, United Bank for Africa, Africa Merchant Bank			
Okpai	Equity: Nigerian National Petroleum Corporation (Nigeria, 60%), Nigerian Agip Oil Company (Italy, 20%), and Phillips Oil Company (USA, 20%) maintained equity since 2001 Debt: 100% equity financed	DN	Y	0
Afam VI	Equity: Nigerian National Petroleum Corporation (Nigeria, 55%), Shell (UK/Netherlands, 30%), Elf (Total) (France, 10%), Agip (Italy, 5%) Debt: 100% equity financed	DN	N	0
Aba Integrated	Equity: Geometric Debt: Senior debt: Diamond Bank (Nigeria) and Stanbic IBTC Bank (Nigeria); Subordinated debt: EIB and EAIF	DN	N	0
Tanzania				
IPTL	Equity: Mechmar (Malaysia, 70%), VIP (Tanzania, 30% in kind); sold to Pan Africa Power Tanzania Ltd (PAP) in 2013 (disputed) Debt: Bank Bumiputra and Sime Bank (Singapore); Standard Chartered Bank, Hong Kong (SCB-HK) bought debt, valued at \$125 million, for \$74 million (in 2005)	DN	Y	1
Songas	Equity: TransCanada sold majority shares to AES (USA) in 1999 and AES sold majority shares to Globeleq (UK) in 2003. All preferred equity shares were converted into "Loan Notes" in June 2009, only common shares remain Debt: IDA (\$120 million), EIB (\$50 million), assumed loans of \$69.2 million from initial TANESCO plant	ICB	Y	2
Mtwara	Equity: Artumas Group Inc. (Canada, 100%), sold shares to Wentworth Group, which in turn sold to TANESCO in 2012 Debt: 100% financed with balance sheet of shareholders	ICB	Y	2
Symbion	Equity: Built by Richmond, sold to Dowans, then to Symbion Debt: equity financed	DN	Y	2
Uganda^b				

(continued on next page)

Table 1 (continued)

Project	Equity partners (country, % of equity held)	Procurement	Contract change Y=yes/N=no	Equity turnover (#) ^a
Bujagali	Equity: Sithe Global (USA, 58%), IPS-AKFED (32%), Government of Uganda (10%) Debt: IFC, EIB, Proparco, KfW, AfDB, FMO, DEG, AfDB, Standard Chartered, ABSA	ICB	N	0
Namanve	Equity: Jacobsen (Norway, 100%) Debt: Norwegian commercial bank and local Ugandan bank, and supported by the Norwegian Agency for Development Cooperation (NORAD)	ICB	N	0
Bugoye	Equity: TrønderEnergi, Norfund (Norway) Debt: EAIIF/FMO	DN	N	0
Mpanga	Equity: South Asia Energy Management Systems (SAEMS) (USA, 100%) Debt: EAIIF, FMO, DEG	DN	N	0
Tororo	Equity: Electro-Maxx (Uganda, 100%) Debt: funded by local Ugandan banks	DN	N	0
Ishasha	Equity: Eco Power Ltd (Sri Lanka, 100%) Debt: Sri Lankan commercial banks	DN	N	0
Buseruka	Equity: Hydromax Limited (Uganda, 100%) Debt: African Preferential Trade Area Bank (PTA), AfDB	DN	N	0

ABSA = South African commercial bank; AfDB = African development Bank; AIIF = African Infrastructure Investment Fund; BWSC = Danish engineering company now owned by Mitsui; CDC = Commonwealth Development Corporation; DEG = German Investment and Development Corporation; DFIs = development finance institutions; DN = direct negotiation; EAIIF = Emerging Africa Infrastructure Fund; EIB = European Investment Bank; EKF = Eksport Kredit Fonden—Danish export credit fund; FMO = Netherlands Development Finance Company; GETFIT = global energy transfer feed-in-tariff; ICB = international competitive bid; IDA = International Development Association; IFC = International Finance Corporation; IFU = Danish Investment Fund for Developing Countries; IPS = Industrial Promotion Services; IPS-AKFED = Industrial Promotion Services Aga Khan Fund for Economic Development; IPTL = Independent Power Tanzania Ltd; KfW = German Development Bank; KP & P = Company registered in the Netherlands to develop the Lake Turkana Wind project; KPLC = Kenya Power and Lighting Company; OPEC = Organization of the Petroleum Exporting Countries; OPIC = Overseas Private Investment Corporation; REFIT = renewable energy feed-in-tariffs; RMB = Rand Merchant Bank; TANESCO = Tanzania Electric Supply Company; YFP = Yinka Folaio Power.

^a Shareholders—particularly those with technical expertise—are often prohibited (by lenders) from selling until after commercial operation.

^b The balance of four Ugandan IPPs (Kilembe Mines aka Mubuku I, Kakira, Kinyara, and Kasese Cobalt aka Mubuku III), not included in the table above, were developed to source electricity to the mining/sugar industries and have evacuated excess power to the national grid. Also not included are the eight GETFITs and two solar ICBs for which financial close was imminent but not yet complete in 2014.

credible and predictable regulatory decisions are necessary for creating certainty around market access, tariffs, and revenues that encourage investment (Eberhard and Gratwick, 2011).

4.3. Planning, procurement & contracting

Apart from the quality of regulation, there are further elements that play an important role in attracting IPP investment. The first set of issues relate to generation planning, procurement and contracting.

4.3.1. Planning

In the course of power sector reform and restructuring, least-cost power planning is often neglected. Typically, this responsibility shifts from the incumbent national utility to the ministry of energy or the regulator, but these institutions seldom have the experience or resources to produce regular and flexible power plans. Additionally, in hybrid power markets it is often unclear who is responsible for generation expansion planning.

The ideal location for this function is the system operator, which can be assigned responsibility not only for short-term, but also medium and long-term supply security. However, this only makes sense if the system operator and transmission utility are unbundled from the state power generation company so that they can plan objectively and fairly for the entire national system. An unbundled system operator can also be responsible for procurement and contracting of new power generation capacity. In several countries in Africa, this function remains within the national utility, in which case the government may exercise political leadership to ensure that the incumbent utility works in the national interest. Whatever the arrangements are, it is critical that the responsible agency be resourced with adequate capacity.

The nature of power sector planning is equally important: it needs to be up-to-date and flexible to ensure security of supply, a least-cost mix of generation plants, and the right combination of exports and imports. Currently, the majority of Sub-Saharan African countries have inadequate planning capacity and end up contracting out this function

to consultants (Eberhard and Gratwick, 2011). All five case study countries have some form of least cost power development plans developed, although in the cases of South Africa, Nigeria and Uganda these plans are severely outdated.

4.3.2. Procurement & contracting

For electricity plans to matter, they need to be translated into timely procurement and well-delineated investment opportunities for the private and public sector. Unfortunately, few African countries have an explicit connection between planning and procurement (Malgas and Eberhard, 2011). In addition, clearly stated criteria for the allocation of investment opportunities between state-owned enterprises (SOEs) and IPPs are lacking.

A key feature of power generation procurement in Africa is the low recourse to competitive bidding, despite the fact that this is frequently enshrined into legislation. A minority of IPPs in SSA, excluding South Africa, have been competitively procured: only 16 competitive tenders versus 34 directly negotiated projects. This is partly due to the first IPPs being procured in reaction to power shortages (in all five case study countries), in contexts where capacity planning was weak. In contrast, competitive tenders require good planning, procurement and contracting frameworks – and often impose higher transaction costs. They have, however, delivered better price outcomes than directly negotiated power projects, even in situations with too few bidders (Fig. 9). Competitive tenders are also more transparent, providing better investment certainty and less chance of corruption.

More than two decades of experience in power procurement in Sub-Saharan Africa have demonstrated that a lack of competition in procuring new generation capacity has extensive drawbacks, ranging from immediate effects on project outcomes (higher prices, unravelling contracts, and so on) to more general effects on the overall governance of the electricity sector and its investment climate. Despite this, we see both forms of procurement used relatively consistently from 1990 to 2014, with no clear move away from or towards either direct negotiations or competitive bidding (Fig. 10).

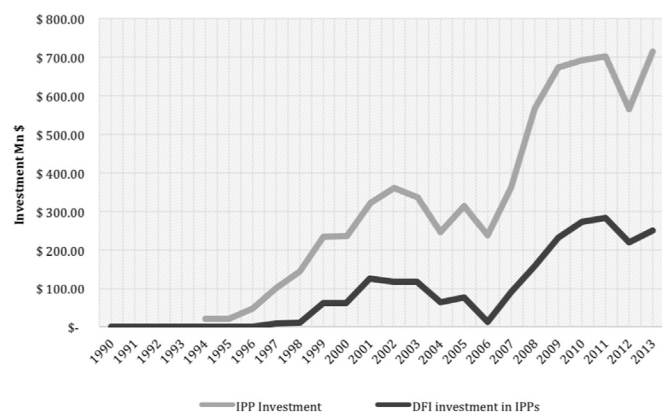


Fig. 7. Total investment by IPPs and by Development Finance Institutions: Sub-Saharan Africa (Excluding South Africa), 1994 – 2014. Note: DFI = development finance institution; IPP = Independent Power project.

In most cases, IPP contracts extend over a long period of time; the typical contract is for 15–30 years. This long time frame is considered both a strength and a weakness. Predictable revenue streams allow equity risk capital to be rewarded, and sponsors can also service debt with long tenors. Conversely, in an environment of power market reform, both parties can encounter problems with fixed long-term take-or-pay contracts if the various conditions under which the contracts are agreed upon change. Governance frameworks, which shape the degree

of predictability and risk in the sector, ultimately impact on investment and development outcomes.

4.4. Risk mitigation

At the crux of the investment conundrum is the security of revenue flows and hence the financial viability of the off-taker – usually the national utility. Many African utilities have poor credit ratings and perform inefficiently (Eberhard and Shkaratan, 2012). Average distribution losses in Sub-Saharan Africa are 23%, compared with the commonly used norm of 10% or less in developed countries. Moreover, average collection rates are only 88.4% compared with the best practice of 100%. Combining the costs of distribution losses and uncollected revenue, expressed as a percentage of utility turnover, provides a measure of a utility's inefficiency. In Africa, this efficiency is equivalent, on average, to 50% of turnover (Eberhard et al., 2011).

At the sector level, governance reforms targeting ownership and shareholder quality, managerial and board autonomy, accounting standards, performance monitoring, outsourcing, labour markets and capital market discipline can critically improve the performance of state-owned utilities. Most utilities in Sub-Saharan Africa meet only about half of the criteria for good governance. At the operational level, practices targeting technical and commercial efficiency will ameliorate the financial standing of a utility in a short period of time (Eberhard and Shkaratan, 2012). Such actions become increasingly important as a utility approaches an IPP transaction, since these directly affect its ability to honour payment obligations.

Given the performance of most state-owned utilities in SSA, credit

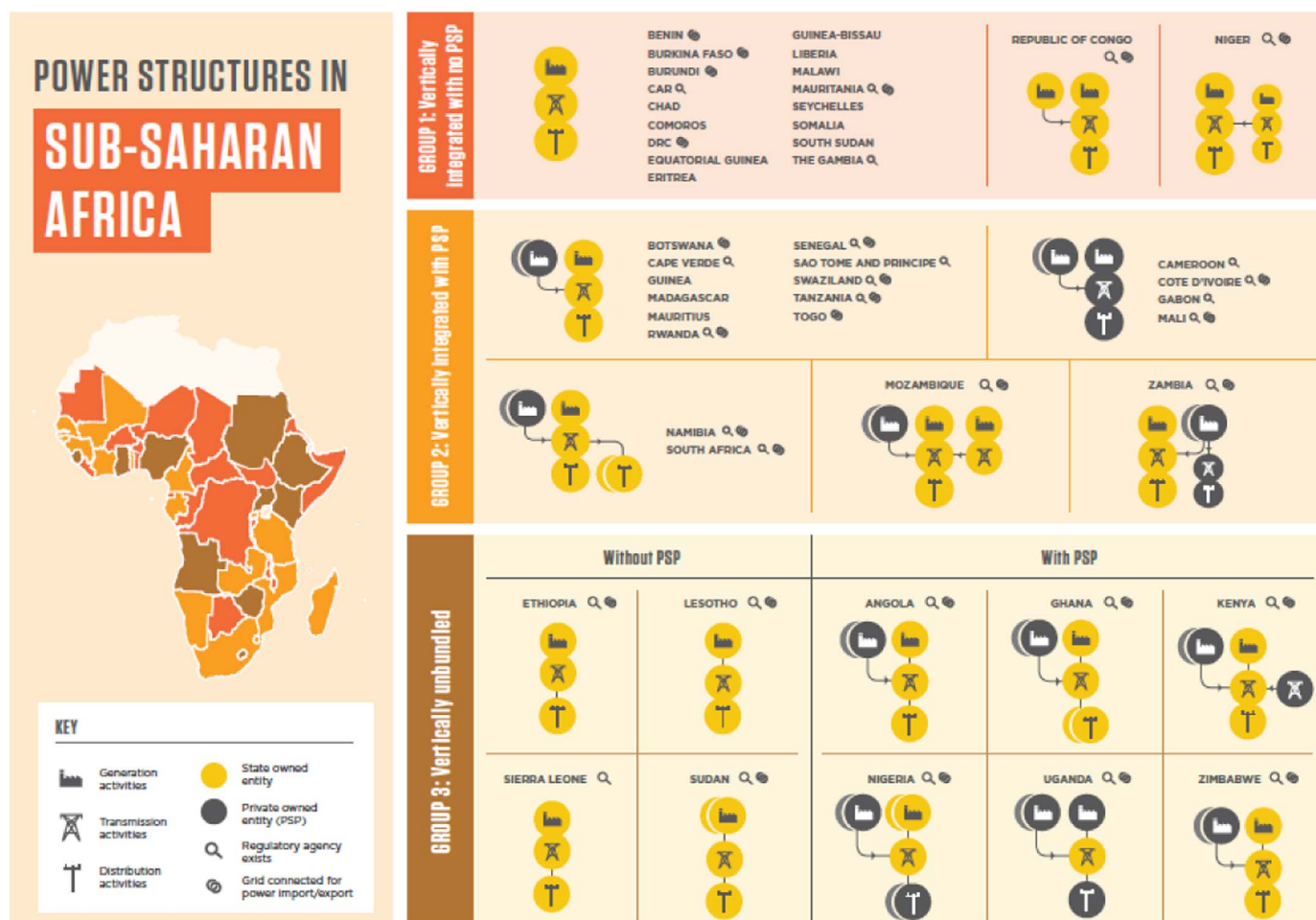


Fig. 8. Electricity sector structures in Sub-Saharan Africa, 2014. Note: includes vertical integration or unbundling of generation (G), transmission (T), and distribution (D) and presence of IPPs. CAR = Central African Republic; DRC = Democratic Republic of Congo; IPP = independent power project; PSP = Private Sector Participation.

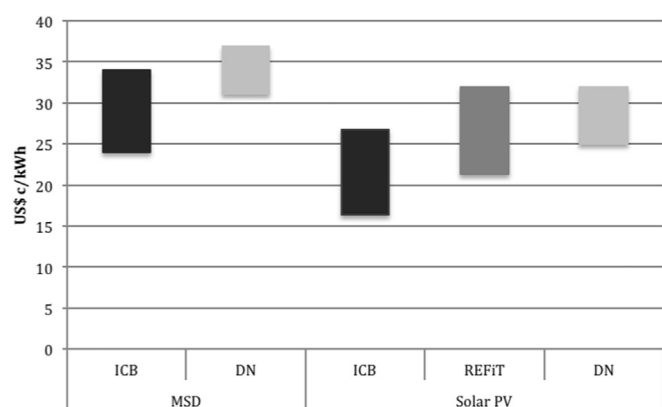


Fig. 9. Comparison of price outcomes for Medium-Speed Diesel and Solar PV in SSA. Note: ICB – International Competitive Bid, DN – Directly Negotiated, REFiT – Renewable Energy Feed-In Tariff, MSD – Medium-Speed Diesel, PV – Photovoltaic.

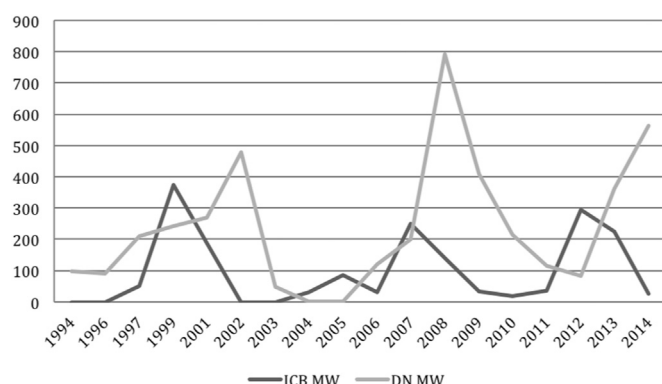


Fig. 10. Competitive tenders vs. directly negotiated projects, excluding South Africa, 1994–2014 (MW). DN = direct negotiation; ICB = international competitive bid; MW = megawatts.

Source: Based on authors' calculations.

enhancement and risk mitigation – including payment and termination risks – are therefore important for attracting IPP investment in SSA. Robust PPAs – most often denominated in US dollars or euros – have become a requirement for new investors seeking to safeguard payment streams. Risks can furthermore be reduced through governance reforms; providing investor certainty around planning, procurement and contracting; through the involvement of development finance institutions; ring-fenced revenue, escrow accounts, letters of comfort and credit, guarantees, put-call options on termination; and political insurance. Our analysis of IPPs shows that significant de-risking is necessary to attract private equity and debt-providers (Eberhard et al., 2016).

DFI involvement, in particular, has played a critical role in bringing IPPs to financial close. DFIs appear to “crowd-in” private investments by their ability to exercise pressure on the host government to honour contracts, as well as their offering risk mitigation products such as guarantees and insurance.^c As a result, we have seen very few IPPs actually unravelling, and a renegotiation of contracts^d (after the PPA was signed) in only eight of the 27 IPPs in Kenya, Nigeria, Uganda and Tanzania.

^c While risk mitigation has been critical to attract private investment to IPPs, in no project have guarantees of any sort been invoked, including projects whose contracts ultimately unraveled.

^d Such changes vary, from a scaling back of project size, to a reconfiguration of the project in its entirety. Further changes have included the rolling back or elimination of certain security arrangements to reduce the financial liability of the state-owned utility. It should be noted that changes occurred in what might be termed the first wave of IPPs in Sub-Saharan Africa – which may signal there has been a learning process.

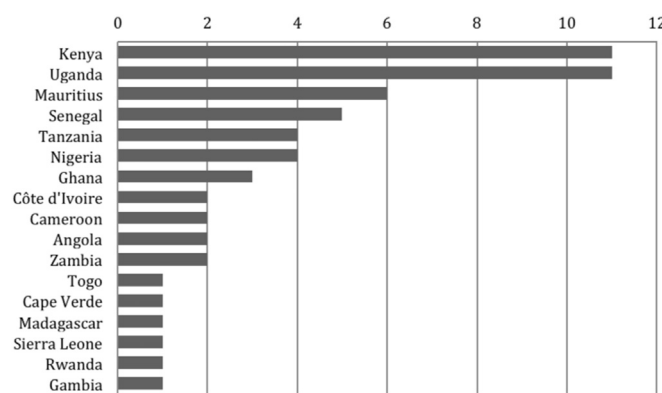


Fig. 11. Number of Independent Power Projects in Sub-Saharan Africa by country (excl. South Africa). 1994–2014.

Source: Compiled by authors, based on utility data, primary sources, and the PPI database.

Of the 47 IPPs that have reached financial close between 1990 and 2015 in these four case study countries, 29 have some form or another of DFI involvement. 18 of these projects have received direct DFI investment totalling US\$ 2 198 million through equity or loans, representing on average 47% of the total investment for these projects. While five IPPs have some form of “foreign” guarantee or risk insurance in place, 12 projects have benefitted from some form of ODA grant (Eberhard et al., 2016). It is thus clear that for the majority of IPPs in the case study countries, DFI's played a significant role, either through the use of risk mitigation products or, primarily, through actually investing in these projects.

4.5. Performance of five case study countries

Fig. 11 provides a visual breakdown of the number of IPPs per country (excluding South Africa). Of the present pool of 58 IPPs that have reached financial close, Kenya and Uganda have the highest number.^e It is noteworthy that three-quarters of the projects in these two countries have closed within the past three years. Thus, more than 50% of the total IPP pool, in terms of number of projects, is concentrated in two countries and is relatively new. The balance developed slowly over the two decades since the first large-scale IPP reached financial close in 1994 in Côte d'Ivoire.

Table 2 below provides a summary of some of the key power sector features in the five case study countries.

Of the five countries, South Africa clearly has the best investment climate, a policy for expanding renewable energy, a power plan linked to a series of competitive tenders, and a set of standardized contracts backed by a sovereign guarantee. The country has an independent regulator, although its decisions have not always been consistent. It could be argued that utility tariffs do not fully reflect costs; nevertheless, the regulator has mandated the full pass-through of IPP costs. The consequence has been a highly successful IPP program where more megawatts and investment have been contracted in four years than in the previous two decades across the rest of Sub-Saharan Africa. Remarkably, this has been achieved within an electricity sector that is dominated by a large state-owned vertically integrated utility that relies mostly on coal and once was not receptive to IPPs.

Kenya has an investment climate that is better than that of neighboring Tanzania and Uganda, as well as Nigeria, and has been able to attract private investment at a lower cost than these countries.

^e Projects included in this tally are all grid-connected IPPs with a capacity of 5MW and greater. A complete list is provided in Appendix 1. While Zimbabwe has three hydro-power IPPs, these projects are all under 5MW and are therefore excluded here.

Table 2

Summary of power sector features in case study countries.

Source: Compiled by authors, based on various primary and secondary source data.

Country	Unbundled utility	Privatized utility	Wholesale competition	Independent regulator	Least-cost power development plan (LCPDP)	Predominant procurement practices
South Africa	No	No	No	Yes	Integrated Resource Plan for 2010–2030 out of date	Competitive
Kenya	Yes	No	No	Yes	LCPDP based on stakeholder consultations	Competitive
Tanzania	No	No	No	Yes	Electricity Supply Reform Strategy and Roadmap, 2014–2025; LCPDP, 2013	Mostly direct negotiations (some previous tenders with limited competition)
Uganda	Yes	Partial ^a	No	Yes	2011 Power Sector Investment Plan not updated	Direct negotiations until advent of GETFIT (hybrid feed-in tariff with competitive tenders)
Nigeria	Yes	Yes	Transitional market	Yes	System operator is mandated to prepare a power master plan but has not been updating it	Direct negotiations

Note.

^a Uganda's main distribution utility is concessioned to a private company, as is the previous state generation utility. Transmission remains public. There are also some small private regional concessions not connected to the main transmission grid. GETFIT = global energy transfer feed-in-tariff.

Its electricity sector has been unbundled, it has an independent regulator, and it once had a clear power-planning process and a competent procurement capability in the Kenya Power and Lighting Company (KPLC), the T & D company. The regulator has helped move tariffs to cost-reflective levels, and the KPLC has been reasonably creditworthy. The consequence is a series of competitive procurements with steadily better price outcomes.

Tanzania on the other hand has a weaker investment climate, some ambivalence around private sector investment, a vertically integrated state-owned utility with technical and financial performance challenges, and poor planning and procurement practice—despite a regulator that seeks to encourage more transparent and competitive procurement. Tanzania has relied more on unsolicited bids and direct negotiations than on competitive tenders. As a result, some IPPs here stand out for their high prices and controversial contracts.

Uganda's recent success has relied less on its overall investment climate and more on a clear power sector structure and a recent competitive tendering program for small renewable energy power plants. With its power sector unbundled, IPPs contract directly with the transmission company, free of conflicts with state-owned generation, and the privately concessioned distribution company is increasingly more effective in reducing losses and improving its financial viability. The dedicated global energy transfer feed-in-tariff (GETFIT) intervention has provided transaction advice and support for running competitive tenders coupled with standardized contracts. It remains to be seen whether this initiative can be sustained in the future.

Nigeria's investment climate is challenging; its previous success with IPPs had less to do with a clear policy framework and more with strong political will at the highest levels. A protracted and torturous power sector reform process—including full unbundling, privatization, and, eventually, competition—has, in the short term, probably made it harder to secure investments in new IPPs. It is hoped that, eventually, the reform process will improve the financial viability of the sector, and the Nigerian Bulk Energy Trader (NBET) will become a dependable and attractive off-taker for IPPs.

This analysis of the case study countries reveals no single or consistent element that guarantees IPP investment. Planning and competitive procurement practices are important; creditworthiness of off-taker utilities is also critical, but policy makers should not lose sight of the broader investment, policy, and regulatory climate.

5. Renewable energy technologies are breaking through

The majority of IPPs in Africa have historically been thermal, mainly

gas or diesel plants. However, we are seeing rapid growth in grid-connected renewable energy capacity, especially wind and solar. The South African Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) is an important illustration of this, with more than 6327 MW of renewable energy contracted in 4 years, across 92 projects, with an investment of more than US\$19 billion (Eberhard et al., 2016, 2014). South Africa has added more renewable energy capacity and IPP projects in this short period of time, than the rest of Sub-Saharan Africa has in more than 20 years. Over four bid rounds, prices fell 46% for wind energy and 71% for solar PV to levels lower than the utility's average cost of supply (Eberhard and Käberger, 2016).

While South Africa clearly leads the region in terms of renewable energy IPPs, it is important to note that this trend is not limited to a single country. Fig. 12 below provides an overview of renewable energy capacity (excluding hydro) from 2006 to 2015 for Sub-Saharan Africa, clearly showing a significant increase in capacity beyond South Africa.

Most of the new RE capacity in the region has been contracted in South Africa, through the competitive procurement program (REIPPPP). Looking beyond South Africa, it is clear that the REIPPPP is having a significant regional impact, with eight Sub-Saharan African countries officially having developed renewable energy auction policies by 2015 (REN21, 2016). This is a significant development given that, apart from South Africa, there were no competitive procurement programmes for renewable energy in the SSA region in

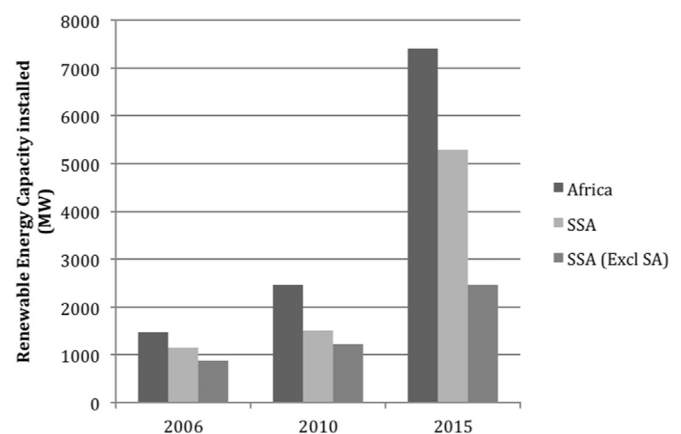


Fig. 12. Renewable Energy Installed Capacity (megawatt) in Africa, 2006 – 2015. Note: Excludes hydropower; SSA = Sub-Saharan Africa; SA = South Africa;.

Source: Compiled by authors based on IRENA Annual Renewable Energy Statistics.

2011 (REN21, 2012).

The Ugandan GETFiT program further illustrates the growth in competitively procured renewable power. In 2013, KfW assisted the Ugandan Regulatory Authority to develop the GETFiT to incentivise new investments, aiming to add 170 MW to the country's 840 MW installed capacity (in 2013) (Meyer et al., 2015). The primary GETFiT mechanism is a grant-based premium payment at the REFiT level to close the gap with the levelised cost of electricity (LCOE) for eligible technologies, namely small hydropower, biomass, bagasse, and solar PV. A competitive tender run in 2014 for four 5 MW solar PV plants delivered an average levelised tariff of US\$ 16.37/kWh – lower than the average retail tariff of US\$ 16.6/kWh in 2013 (Meyer et al., 2015).

Furthermore, Zambia's very recent experience with solar PV IPPs could be a technological game-changer for the continent: through the IFC's Scaling Solar program, two 50 MW solar PV projects were put out to tender. The lowest winning bid came in at US\$ 6.02/kWh – a new record for Africa. The tariffs are fixed (non-indexed) for the 25-year contract period (International Finance Corporation, 2016).

These are but two examples outside of South Africa in the region. Scaling Solar is set to soon being rolled out in Madagascar, Senegal and Ethiopia (Eckhouse, 2016; Sibanda and MacInnis, 2016). Ghana and Burkina Faso concluded RE auction rounds in 2016, and Botswana, Namibia, Ethiopia and Senegal launched pre-qualification or proposal submission stages for their respective auction programmes in the same year. Kenya and Nigeria are additionally in the process of developing auction programmes and more countries are soon to follow. While the majority of renewable energy projects on the continent contracted in the past 3 years remain based on feed-in tariffs or directly negotiated deals, this surge in competitive procurement programmes seems to indicate a noteworthy shift.

Given these trends, the renewable energy sector is set to grow and diversify on the continent, with major additions in especially wind, solar PV and geothermal generation projected to come online in the next few years (Marks et al., 2016; Quitzow et al., 2016).

6. Conclusion and policy implications

As demonstrated by the data on investment trends reported in this paper, IPPs make an important and growing contribution to meeting Sub-Saharan Africa's power needs. These projects are concentrated in a few countries, but there is scope to widen private investment across the continent. Current experience with IPPs on the continent offers important lessons for how this can be done (Eberhard et al., 2016).

Previously published research on critical success factor for IPPs includes a range of country specific factors (investment climate, power sector policies and regulation, effective planning and competitive procurement) and project specific factors (well structured, bankable projects with experienced sponsors and debt providers, robust PPAs, risk mitigation and security measures, and strategic project management) (Eberhard and Gratwick, 2011).

An analysis of the case study countries that have had the most success in attracting IPPs in Sub-Saharan Africa, suggests that while all of the above success factors are important, more emphasis needs to be placed on dynamic power planning, competitive procurement practices and adequate contracting capacity. Power planning cannot be neglected and is an essential first step towards greater private sector power investment. Sound planning means that countries are able to project future electricity demand correctly, decide on least-cost supply options, and anticipate how long it would take to procure, finance, and build the required generation capacity. Planning tools must be updated frequently and new build opportunities should be allocated between public and private generators on the basis of clear criteria. Finally, there must be an explicit link between planning and the timely

initiation of international competitive bids or auctions for long-term contracts for new power. Without these policy tools, capacity and reforms, a country is unlikely to see new privately financed power generation capacity.

IPP investments in Africa will rely on long-term contracts with off-takers. Competitive tenders for these long-term contracts – also called competition for the market – more often than not result in better investment and price outcomes than feed-in-tariffs or directly negotiated projects. This is especially true for standardised technologies such as renewable energy, diesel-generation and heavy-fuel oil generators. Competitive tenders are also more transparent, leading to less corruption. The use of standard contracts in competitive tenders furthermore result in a fair allocation of risk. And projects are more likely to move to financial close, construction and commercial operation. Competitive procurement has also been shown to lead to the development of a pipeline of bankable projects, especially for renewable energy.

More competitive tenders should be run in a greater number of countries, both for standard thermal technologies as well as for other technologies and contexts where competition is possible. While the costs of running a competitive procurement program might be higher than those involved in directly negotiated projects, these are justified by the lower prices achieved. If directly negotiated projects are to be used, governments need to build capacity to ensure that they offer value for money, and that the project developers have the required technical and financial capabilities.

IPP contracts should be undertaken with financially viable off-takers, whether these be utilities or large customers. Secure revenue flows are essential for ensuring the bankability of IPPs, most of which are project-financed. While credit-enhancement and security measures can mitigate some of this risk, the ultimate goal for host governments and development partners should be the improved performance of off-takers (in most cases utilities) as sustainable solution.

The case studies demonstrate that Development Finance Institutions are playing an important role in IPP financing and risk mitigation. The mere presence of these institutions seems to attract private sector interest, particularly due to their ability to exercise pressure on host governments to honour commitments. However, care should still be taken to ensure that where adequate commercial financing appetite exists, DFIs are not taking on more than their appropriate share of financing volume.

A further notable conclusion of the paper is that IPPs have been developed in countries with very different power sector structures and levels of reform. This conclusion does not imply that power sector reform is unimportant but it does underline the relevance of additional factors, noted above, in accelerating investment in IPPs. In developing countries in Sub-Saharan Africa, some reforms may be more important than others. Clearly full wholesale or retail competition is not a pre-condition for private investment in power. But clear policy and effective regulatory frameworks are important for securing market entry, revenue flows and contract prices. And the separation of state-owned generation companies from the main transmission system (as in Kenya) provides a level-playing for IPPs and more investment certainty and confidence.

Finally, we have noted the extent to which renewable energy IPPs are breaking through and can be competitively procured. The significant renewable energy price reductions evidenced by the latest auctions clearly show that renewable energy can compete with fossil-fuel based sources and in many countries solar and wind energy are now the cheapest sources of power. This has significant implications for how the continent's relatively undeveloped power systems are designed. Most African countries have excellent solar resources, and many have good wind resources as well. But solar and wind energy are also variable and

need to be complemented with flexible resources. This calls for a re-evaluation of the role of utilities and power markets in balancing the system and contracting appropriate back-up and auxiliary services.

In conclusion, investment in African IPPs is growing, but not fast enough. Africa does not have sufficient power. All sources of investment need to be encouraged. For IPPs to flourish, Africa needs dynamic, least-cost planning, linked to the timely initiation of the competitive procurement of new generation capacity. This must be accompanied by the building of effective regulatory capacity that encourages the distribution utilities that purchase power to improve their performance and prospects for financial sustainability – and to

widen access to electricity. IPP investment require appropriate risk mitigation, often helped by the participation of development finance institutions. Renewable energy are breaking through and competitive prices are being achieved in auctions for new power. Such efforts promise to promote sustainable economic and social development across the continent.

Acknowledgements

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Appendix A. Independent Power Projects in Sub-Saharan Africa

See: [Tables A1–A18](#).

Table A1

IPP investments in Angola by project.

Project Information	Project Name Chicapa Hydroelectric Plant	Project Name2 Biocom (Malange)
Capacity (MW)	16	30
Technology	Hydro, Small (< 20 MW)	Waste/bagasse
Total Investment (US\$ Million)	45.0	89.8
Year of financial closure	2003	2014
COD	2008	
Project status	Operational	Operational
Procurement method	Direct Negotiation (DN)	DN
Number of bids		
Contract period	40	
Contract type	Build-Operate-Transfer (BOT)	
Sponsors/Developer	ALROSA Co. Ltd. (The Almazay Rossii-Sakha Company) (55%/Russian Federation)	
EPC		
Fuel arrangement		
Debt-equity Ratio		
Local shareholder equity (Entity, US\$ Million)		
Foreign shareholder equity (Entity, US\$ Million)		
DFI Agency and financing method		
Total DFI financing (US\$ Million)		
ODA Grants (US\$ Million)		
Local credit enhancements & security arrangements		
Foreign credit enhancements & security arrangements		

Table A2

IPP investments in Cameroon by project.

Project Information	Project Name Dibamba Power Plant	Project Name2 Kribi Power Plant
Capacity (MW)	88	216
Technology	HFO/MSD	CCGT
Total Investment (US\$ Million)	126.0	342.0
Year of financial closure	2009	2010
COD	2009	2013
Project status	Operational	Operational
Procurement method	DN	DN
Number of bids		
Contract period	20	20
Contract type	BOT	BOT
Sponsors/Developer	AES Corporation (56%/United States), Republic of Cameroon (44%)	KPDC was 56% owned by AES, with the remaining 44% in the hands of the Cameroon government, built by Finland's Wartsila, running on natural gas from the off-shore Sanaga-South field operated by Cameroon's state oil company, SNH, and independent producer Perenco - the first major commercial development of Cameroon's substantial gas reserves. In Nov 2013, AES announced it would sell its stake in Cameroon to Actis (Globeleq parent company), a global pan-emerging market investor, for \$220 million of net equity proceeds. Sale was completed in 2014.

(continued on next page)

Table A2 (continued)

Project Information	Project Name Dibamba Power Plant	Project Name2 Kribi Power Plant
EPC		Wartsila
Fuel arrangement	HFO/Tolling agreement with AES Sonel as Toller	A gas supply agreement has been signed with a state-owned gas supplier.
Debt-equity Ratio		75/25
Local shareholder equity (Entity, US \$ Million)		
Foreign shareholder equity (Entity, US\$ Million)		
DFI Agency and financing method	IFC (Loan/\$31 Million/2010), AFDB (Loan/\$31 Million/2010), (FMO Loan/\$31 Million/2010)	AFDB (Loan/\$57 Million/2011), EIB (Loan/\$41 Million/2012), Other (Loan/\$23 Million/2012), IDA (Guarantee/\$82 Million/2012), IFC (Loan/\$77 Million/2012)
Total DFI financing (US\$ Million)	93.0	198.0
ODA Grants (US\$ Million)		
Local credit enhancements & security arrangements		Sovereign guarantee
Foreign credit enhancements & security arrangements	Typical project finance security agreements implemented but details not made public	WB PRG (enabled local bank participation)

Table A3

IPP Investments in Cape Verde by Project.

Project Information	Project Name Electra Cabeolica Wind Project
Capacity (MW)	25.5
Technology	Wind, Onshore
Total Investment (US\$ Million)	80.0
Year of financial closure	2010
COD	2010
Project status	Operational
Procurement method	
Number of bids	
Contract period	20
Contract type	Build-Own-Operate (BOO)
Sponsors/Developer	Electra (Cape Verde), Africa Finance Corporation (Nigeria)
EPC	
Fuel arrangement	
Debt-equity Ratio	
Local shareholder equity (Entity, US\$ Million)	
Foreign shareholder equity (Entity, US \$ Million)	
DFI Agency and financing method	EIB (Loan/\$39 Million/2010), AFDB (Loan/\$19 Million/2010)
Total DFI financing (US\$ Million)	58.0
ODA Grants (US\$ Million)	
Local credit enhancements & security arrangements	Variable government payments
Foreign credit enhancements & security arrangements	

Table A4
IPP investments in Côte d'Ivoire by project.

Project Information	Project Name	Project Name2	Project Name3	Project Name4	Project Name5	Project Name6
	Compagnie Ivoirienne de Production d'Electricite (CIPREL)	Compagnie Ivoirienne de Production d'Electricite (CIPREL)	Azito Power Project	Compagnie Ivoirienne de Production d'Electricite (CIPREL)	Azito Power Project	Compagnie Ivoirienne de Production d'Electricite (CIPREL)
Capacity (MW)	99	111	288	111	146	111
Technology	OCGT	OCGT	OCGT	OCGT	OCGT+CCGT	OCGT+CCGT
Total Investment (US\$ Million)	108.0	134.0	223.0	134.0	207.0	134.0
Year of financial closure	1994	1997	1999	2009	2013	2013
COD	1995		2000			
Project status	Operational, Planning/reached financial closure	Operational, Planning/reached financial closure	Operational	Operational	Under Construction	Reached financial close
Procurement method	DN	DN	International Competitive Bid (ICB)	DN	ICB	DN
Number of bids	19		24			
Contract period			BOOT			
Contract type	Build-Own-Operate Transfer (BOOT)					
Sponsors/Developer	SAUR International, with 88% (Joint Venture (JV) between French SAUR Group owned by Bouygues, 65% and EDF, 35%) BOAD, PROPARGO and IFC holding the remaining 12%; in 2005 all shares sold to Bouygues (France, 98%), except BOAD (2%)	SAUR International, with 88% (JV between French SAUR Group owned by Bouygues, 65% and EDF, 35%) BOAD, PROPARGO and IFC holding the remaining 12%; in 2005 all shares sold to Bouygues (France, 98%), except BOAD (2%)	Globeleq (77% United Kingdom), Aga Khan Fund (Switzerland)	SAUR International, with 88% (JV between French SAUR Group owned by Bouygues, 65% and EDF, 35%) BOAD, PROPARGO and IFC holding the remaining 12%; in 2005 all shares sold to Bouygues (France, 98%), except BOAD (2%)	SAUR International, with 88% (JV between French SAUR Group owned by Bouygues, 65% and EDF, 35%) BOAD, PROPARGO and IFC holding the remaining 12%; in 2005 all shares sold to Bouygues (France, 98%), except BOAD (2%)	SAUR International, with 88% (JV between French SAUR Group owned by Bouygues, 65% and EDF, 35%) BOAD, PROPARGO and IFC holding the remaining 12%; in 2005 all shares sold to Bouygues (France, 98%), except BOAD (2%)
EPC						
Fuel arrangement	Government procures fuel	Government procures fuel	Government procures fuel	Government procures fuel	Government procures fuel	Government procures fuel
Debt-equity Ratio						
Local shareholder equity (Entity, US\$ Million)						
Foreign shareholder equity (Entity, US\$ Million)						
DFI Agency and financing method	BOAD (Loan/\$9 Million/1994), IFC (Loan/\$18 Million/1995), IFC (Equity/\$1 Million/1995), IBRD (Loan/\$80 Million/1995)		AFDB (Loan/\$14 Million/1998), IDA (Guarantee/\$30 Million/1999), IFC (Loan/\$41 Million/1999), IFC (Syndication/\$31 Million/1999)			IFC, AfDB and PROPARGO
Total DFI financing (US\$ Million)	108.0	-	116.0	-	-	-
ODA Grants (US\$ Million)						
Local credit enhancements & security arrangements			Sovereign guarantee, Escrow account equivalent to 1 month capacity charge			
Foreign credit enhancements & security arrangements			World Bank PRG			

Table A5

IPP investments in Gambia by project.

Project Information	Project Name Brikama
Capacity (MW)	25
Technology	HFO+MSD/HFO
Total Investment (US\$ Million)	36.2
Year of financial closure	2005
COD	2006
Project status	Operational
Procurement method	
Number of bids	
Contract period	
Contract type	
Sponsors/Developer	Global Electric Company (GEG)
EPC	
Fuel arrangement	
Debt-equity Ratio	
Local shareholder equity (Entity, US\$ Million)	
Foreign shareholder equity (Entity, US\$ Million)	
DFI Agency and financing method	
Total DFI financing (US\$ Million)	–
ODA Grants (US\$ Million)	
Local credit enhancements & security arrangements	
Foreign credit enhancements & security arrangements	

Table A6
IPP investments in Ghana by project.

Project Information		Project Name	Project Name2	Project Name3	Project Name4	Project Name5
		Takoradi 2	Sunon-Asogli Power Plant	CENT Energy	Takoradi 2	Kpone IPP
Capacity (MW)		220	200	126	110	350
Technology		OCGT/CCGT	OCGT+CCGT	OCGT+CCGT	OCGT+CCGT	CCGT
Total Investment (US\$ Million)		110.0	200.0	140.0	330.0	900.0
Year of financial closure		1999	2007	2009	2013	2014
COD		2000	2011	2012	2014x	2017
Project status		Operational	Operational	Operational	Under construction	Financial close
Procurement method		DN	DN	DN	DN	
Number of bids						
Contract period		25				
Contract type		BOOT	BOO	BOO		
Sponsors/Developer		CMS (USA, 90%), VRA (Ghana, 10%), CMS sold shares to TAQA (UAE, 90%) in 2007	Shenzhen Electric (60%/China), China Africa Development Fund (CADfund) (40%/China)	Gecad (100%/United States)	CMS (USA, 90%), VRA (Ghana, 10%), CMS sold shares to TAQA (UAE, 90%) in 2007	Africa Finance Corporation (AFC) (31.85%), CenPower Holdings Limited (21%), a consortium of Ghanaian investors, Sumitomo Corporation (28%), Mercury Power (15%), and FMO (4.15%)
EPC		Mitsui & Co (Japan) and KEPCO E&C (Korea), Government procures fuel	Interim fuel agreement for access to WAGP gas		Mitsui & Co (Japan) and KEPCO E&C (Korea), Government procures fuel	
Fuel arrangement						72/28
Debt-equity Ratio						
Local shareholder equity (Entity, US\$ Million)			Local strategic investor, Togbe Afede XIV			
Foreign shareholder equity (Entity, US\$ Million)						
DFI Agency and financing method		IFC (Loan/\$60 Million/2004)		Other (Loan/\$67 Million/2008), Other (Quasi-equity/\$10 Million/2008), AFDB (Loan/\$32 Million/2011)	IFC, and a consortium of international development finance institutions led by FMO. The lenders participating in the consortium include the African Development Bank, Deutsche Investitions-und Entwicklungsgesellschaft, Emerging Africa Infrastructure Fund, ICF-Debt Pool and Proparco. The Opec Fund for International Development and the Canada Climate Change Program are participating alongside IFC	FMO (equity, \$10.3 million), DBSA (loan, \$53 million), OFID (loan, \$7 million), EAIIF (\$25 million), FMO (loan, \$24 million), and others
Total DFI financing (US\$ Million)		60.0	–	109.0	347.5	207.0
ODA Grants (US\$ Million)						
Local credit enhancements & security arrangements		Sovereign guarantee (phase 1), US\$ 3 million Letter of Credit provided by government (phase 1)	Variable government payments			
Foreign credit enhancements & security arrangements						

Table A7
A. IPP investments in Kenya by project.

Project Information		Project Name	Project Name2	Project Name3	Project Name4	Project Name5	Project Name6
		Mombasa Barge-Mounted Power Project/Westmont	Iberafrika Power Ltd.	Kipevu II/Tsavo	Ormat Olkaria III Geothermal Power Plant, OrPower4 (phase 1 and 2 and 3)	Iberafrika Power Ltd.	Mumias Power Plant
Capacity (MW)	46	44		75	13	12	26
Technology	OCGT	MSD/HFO		MSD/HFO	Geothermal	MSD/HFO	Waste
Total Investment (US\$ Million)	65.0	50.3		86.0	105.0	13.7	50.0
Year of financial closure	1996	1996		1999	1999	1999	2008
COD	1997	1997		2001	2000, 2009	2000	2009
Project status	Concluded	Operational		Operational	Operational	operational	Operational
Procurement method	DN	DN		ICB	ICB	DN	DN
Number of bids	7	7		3	2		
Contract period	BOO	BOO		20	20	15	
Contract type	Westmont Ltd. (Malaysia)	BOO		BOO	BOO	BOO	BOO
Sponsors/Developer		Union Fenosa (Spain, 80%), KPLC Pension Fund (Kenya, 20%) since 1997		Chnergy & IPS jointly owned 49, 9%, Chnergy sold to Duke Energy in 2005, CDC/Globeleq (UK, 30%), Wartsila (Finland, 15%), IFC (5%) retain remaining shares since 2000	Ormat Turbines Ltd (100%/Israel)	Union Fenosa (Spain, 80%), KPLC Pension Fund (Kenya, 20%) since 1997	Mumias Sugar Company Limited (100%/Kenya)
EPC							
Fuel arrangement	Originally Westmont to procure fuel and then pass through to utility, however, following dispute with fuel supplier about taxes after the first year of operation, utility took over procurement	Iberafrika buys fuel and passes cost through to KPLC based on the units generated and specific consumption parameters agreed on in the PPA	Tsavo buys fuel and passes cost through to KPLC based on the units generated and specific consumption parameters agreed on in the PPA	The only fuel arrangement per se is that OrPower4 was granted a Geothermal Resource License from the government, to which it pays a royalty of sorts (of US\$ 0.04/kWh or USc. 4/kWh)	Iberafrika buys fuel and passes cost through to KPLC based on the units generated and specific consumption parameters agreed on in the PPA		
Debt-equity Ratio	72/28	72/28	78/22				
Local shareholder equity (Entity, US\$ Million)	KPLC Staff Pension Fund (US \$9.4 in direct loans and guaranteed US\$5 million through a local Kenyan bank).	KPLC Staff Pension Fund (US \$9.4 in direct loans and guaranteed US\$5 million through a local Kenyan bank).	9.45				
Foreign shareholder equity (Entity, US\$ Million)	Union Fenosa (Spain) (US\$12.7 million in direct loans and guaranteed US\$20 million);	Union Fenosa (Spain) (US\$12.7 million in direct loans and guaranteed US\$20 million);	9.48		Ormat (100%) since 1998 - UNTIL 2008		
DFI Agency and financing method							
				IFC (Loan/\$18 Million/2000), IFC (Equity/\$2 Million/2000), IFC (Quasi-equity/\$3 Million/2000), IFC (Syndication/\$24 Million/2000), IFC (Risk management/\$2 Million/2001), CDC own account (US\$13 million); DEG own account (€11 million), DEG syndicated (€2 million)	MIGA (Guarantee/\$49 Million/2000), MIGA (Guarantee/\$70 Million/2002), MIGA (Guarantee/\$89 Million/2009), MIGA (Guarantee/\$110 Million/2011)		
Total DFI financing (US\$ Million)	-	-	82.0				
ODA Grants (US\$ Million)	-	-					
Local credit enhancements & security arrangements	An advance payment cash deposit initially, but Iberafrika presently has no payment security	An advance payment cash deposit initially, but Iberafrika presently has no payment security	Letter of Comfort provided by government, escrow account, equivalent to 1 month capacity charge, and a stand-by Letter of Credit, equivalent to 3 months billing	A stand-by Letter of Credit, covering several months billing (although only finalized at end-2006)	An advance payment cash deposit initially, but Iberafrika presently has no payment security		Payment Guarantee
Foreign credit enhancements & security arrangements				MIGA guarantee			

Table A7
B. IPP investments in Kenya by project

Project Information	Project Name7	Project Name8	Project Name9	Project Name10	Project Name11	Project Name12
	Rabai Power Plant	Ormat Olkaria III Geothermal Power Plant, OrPower4 (phase 1 and 2 and 3)	Iberafica Power Ltd.	Ormat Olkaria III Geothermal Power Plant, OrPower4 (phase 1 and 2 and 3)	Triumph HFO Power Plant	Thika Thermal Power Project
Capacity (MW)	90	35	52.5	36	83	87
Technology	MSD/HFO & steam cycle	Geothermal	MSD/HFO	Geothermal	MSD/HFO	MSD/HFO
Total Investment (US\$ Million)	155.0	128.7	59.9	126.2	140.0	144.0
Year of financial closure	2008	2009	2009	2011	2012	2012
COD	2010	2009	2009	2013	2015	2013
Project status	Operational	Operational	operational	Operational	Construction	Operational
Procurement method	ICB	DN	DN	DN	ICB	ICB
Number of bids	4				5	9
Contract period	20		25		20	20
Contract type	BOOT	BOO	BOO	BOO	BOO	BOO
Sponsors/Developer	Aldyeh: 34.%, BWSC (Danish, but owned by Mitsui of Japan); 25.5%, FMO: 20%, IFU (Danish bilateral lender): 20%	Ormat Turbines Ltd (100%/Israel)	Union Fenosa (Spain, 80%), KPLC Pension Fund (Kenya, 20%) since 1997	Ormat Turbines Ltd (100%/Israel)	Broad Holding (Kenya), Interpel Investments (Kenya), Tafeflex (Kenya), Southern Inter-trade (Kenya)	Melec Powergen (part of Maletec grp) (90%/Lebanon)
EPC	BWSC Co-developer, sponsor, shareholder, EPC contractor and O & M contractor				XJ International Engineering Company (wholly owned subsidiary of State Grid Corporation of China)	MAN Diesel (Germany) and Matelec Grp
Fuel arrangement	Fuel Supply Agreement with Kenol of Kenya		Iberafica buys fuel and passes cost through to KPLC based on the units generated and specific consumption parameters agreed on in the PPA			
Debt-equity Ratio	75/25				74/26	75/25
Local shareholder equity (Entity, US\$ Million)						
Foreign shareholder equity (Entity, US\$ Million)						
DFI Agency and financing method		Ormat (100%) since 1998 - UNTIL 2008		Ormat (100%) since 1998 - UNTIL 2008		
	Other (Loan/\$126 Million/2008), DEG: 15%, FMO: 25%, EAF: 25%, Proparco: 25%, EFP (European Financing Partners): 10%	MIGA (Guarantee/\$49 Million/2000), MIGA (Guarantee/\$70 Million/2002), MIGA (Guarantee/\$89 Million/2009), EIB (Loan/\$155 Million/2010), MIGA (Guarantee/		MIGA (Guarantee/\$49 Million/2000), MIGA (Guarantee/\$70 Million/2002), MIGA (Guarantee/\$89 Million/2009), EIB (Loan/\$155 Million/2010), MIGA (Guarantee/	MIGA (Guarantee/\$12 Million/2012), IDA (Guarantee/\$45 Million/2012)	AFDB (Loan/Euro 28 Million/2012), IFC (Loan/Euro28 Million/2012), IDA (Guarantee/\$45 Million/2012), MIGA (Guarantee/\$62

(continued on next page)

Table A7 (continued)

Project Information	Project Name7	Project Name8	Project Name9	Project Name10	Project Name11	Project Name12
	Rabai Power Plant	Ormat Olkaria III Geothermal Power Plant, OrPower4 (phase 1 and 2 and 3)	Iberafrika Power Ltd.	Ormat Olkaria III Geothermal Power Plant, OrPower4 (phase 1 and 2 and 3)	Triumph HFO Power Plant	Thika Thermal Power Project
Total DFI financing (US\$ Million)	126.0	\$110 Million/2011)	–	\$110 Million/2011)	–	Million/2012)
ODA Grants (US\$ Million)	–	–	–	–	–	–
Local credit enhancements & security arrangements	Support Letter from GoK (covers political risk but falls short of being an outright guarantee). KPLC issued a letter of credit equivalent to 5 months of capacity (debt service, fixed costs and equity returns) payments and 2 months of fuel payments	Letter of Comfort provided by government, escrow account, equivalent to 1 month capacity charge, and a stand-by Letter of Credit, equivalent to 3 months billing	An advance payment cash deposit initially, but Iberafrika presently has no payment security	Letter of Comfort provided by government, escrow account, equivalent to 1 month capacity charge, and a stand-by Letter of Credit, equivalent to 3 months billing		
Foreign credit enhancements & security arrangements					PRG	PRG

Table A7

C: IPP investments in Kenya by project.

Project Information	Project Name13	Project Name14	Project Name15	Project Name16
	Kinangop greenfield wind project	Gulf Power	Lake Turkana Wind Power	Ormat Olkaria III Geothermal Power Plant, OrPower4 (phase 1 and 2 and 3)
Capacity (MW)	60	80	300	26
Technology	Wind, Onshore	MSD/HFO	Wind, Onshore	Geothermal
Total Investment (US\$ Million)	150.0	108.0	861.1	91.1
Year of financial closure	2013	2013	2014	2014
COD	Delayed	2014	2017	2014
Project status	Construction/stalled	Operational	Financial close	Operational
Procurement method	REFIT	ICB	DN	DN
Number of bids		5		
Contract period		20	20	
Contract type	BOO	BOO	BOO	BOO
Sponsors/Developer	Aeolus Kenya, AIIF2, which became involved in the project in 2012 to assist the developer Aeolus Kenya to	Consortium of local investors, namely Gulf Energy Ltd and Noora Power Ltd	KP & P Africa BV, a group of Dutch entrepreneurs, with Aldwych International as co-developers	

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Table A7 (continued)

Project Information	Project Name13	Project Name14	Project Name15	Project Name16
	Kinangop greenfield wind project	Gulf Power	Lake Turkana Wind Power	Ormat Olkaria III Geothermal Power Plant, OrPower4 (phase 1 and 2 and 3)
EPC	conclude all material contracts and deliver a bankable project, is the majority owner of the project company Kinangop Wind Park (KWP), while Norfund held the remaining interest.			
Fuel arrangement				
Debt-equity Ratio				
Local shareholder equity (Entity, US\$ Million)				
Foreign shareholder equity (Entity, US\$ Million)				
DFI Agency and financing method	About three-quarters of the 80 million-euro project will be debt-financed. IFC, the OPEC Fund for International Development and Standard Bank Group Ltd. are each lending 20 million euros (\$26 million). \$32 million of equity investments and \$76 million in long-term debt financing. The debt portion consists of IFC A Loan, and commercial lending through IFC B Loan and OPEC Fund for International Development (OFID)			
Total DFI financing (US\$ Million)	–	52.0	595.8	–
ODA Grants (US\$ Million)	–	–	–	–
Local credit enhancements & security arrangements			GOK letter of support	
Foreign credit enhancements & security arrangements		IDA guarantee, MIGA	EKF (Danish export credit agency) to guarantee approx billion Danish kroner in total to: EIB and AfDB	

Table A8

IPP investments in Madagascar by project.

Project Information	Project Name
	Hydelec Madagascar S.A.
Capacity (MW)	15
Technology	Hydro, Small (< 50 MW)
Total Investment (US\$ Million)	17.8
Year of financial closure	2007
COD	2008
Project status	Operational
Procurement method	
Number of bids	
Contract period	15
Contract type	BOT
Sponsors/Developer	Hydelec Madagascar (100%/Madagascar)
EPC	
Fuel arrangement	
Debt-equity Ratio	
Local shareholder equity (Entity, US \$ Million)	
Foreign shareholder equity (Entity, US\$ Million)	
DFI Agency and financing method	AFDB (Loan/\$9 Million/2007), MIGA (Guarantee/\$20 Million/2008)
Total DFI financing (US\$ Million)	9.0
ODA Grants (US\$ Million)	
Local credit enhancements & security arrangements	
Foreign credit enhancements & security arrangements	

Table A9
IPP investments in Mauritius by project.

Project Information	Project Name	Project Name2	Project Name3	Project Name4	Project Name5	Project Name6
	Deep River Beau Champ aka Consolidated energy limited	FUEL power plant	Belle Vue Power Plant	St. Aubin Power Project aka Compagnie thermique du Sud	Compagnie thermique de Savannah	Medine
Capacity (MW)	28.4	36.7	71.2	32.5	90	13
Technology	Waste/bagasse	Waste/bagasse	Coal/bagasse	Waste/bagasse	OCGT/COGT	Waste/bagasse
Total Investment (US\$ Million)	85.0	109.7	109.3	95.2	81.5	38.9
Year of financial closure	1997	1998	1998	2004	2005	1994–2011
COD						
Project status	Operational	Operational	Operational	Operational	Operational	Operational
Procurement method				ICB		
Number of bids						
Contract period		20	20	20		
Contract type	BOO	BOO	BOO	BOO		
Sponsors/Developer	Sugar Investment Trust (10%/Mauritius)	Sugar Investment Trust (20%/Mauritius)	Harel Freres (51%/Mauritius), Sugar Investment Trust (14%/Mauritius), SODEC (27%/France)	Sugar Investment Trust (15%/Mauritius), Mon Tresor Mon Desert (19%/Mauritius), Savannah Sugar Estates (15%/Mauritius), Societe Union St Aubin (15%/Mauritius), Sechillienne-SIDEC (25%/France)		
EPC						
Fuel arrangement						
Debt-equity Ratio						
Local shareholder equity (Entity, US\$ Million)						
Foreign shareholder equity (Entity, US\$ Million)						
DFI Agency and financing method			EIB (Loan/\$17 Million/1998)			
Total DFI financing (US\$ Million)	-	-	17.0	-	-	-
ODA Grants (US\$ Million)						
Local credit enhancements & security arrangements						
Foreign credit enhancements & security arrangements						

Table A10
IPP Investments in Nigeria by Project.

Project Information	Project Name	Project Name2	Project Name3	Project Name4	Project Name5
	AES Nigeria Barge Limited	Okpai Independent Power Project	Afam Power Project	Azura	Aba Integrated (embedded)
Capacity (MW)	270	480	630	450	141
Technology	OCGT/CCGT	OCGT/CCGT	OCGT/CCGT	OCGT	OCGT
Total Investment (US\$ Million)	240.0	462.0	540.0	895.0	460.0
Year of financial closure	2001	2002	2008	2015	2013
COD	2001	2005	2008	2016x	2013
Project status	Operational	Operational	Operational	Financial close expected	Operational
Procurement method	DN	Unsolicited proposals	DN	DN	DN
Number of bids	1	1	8	8	
Contract period	13	20	20	20	
Contract type	BOO	BOO	BOO	BOO	
Sponsors/Developer	Euron (USA, 100%) sold to AES (95%) and YFP (Nigeria, 5%) in 2000	Nigerian National Petroleum Corporation (Nigeria, 60%), Nigerian Agip Oil Company (Italy, 20%, with Agip owned by ENI since 2003), and Phillips Oil Company (USA, 20%) maintained equity since 2001	NNPC (Nigeria, 55%), Shell (UK/Netherlands, 30%), Elf (Total) (France, 10%), Agip (Italy, 5%)	Aldwych International, Africa Infrastructure Investment Fund (AIIM), and Asset and Resource Management (ARM) in conjunction with the government of Edo state which has about five per cent equity stake in the project	Geometric
EPC				Siemens and Julius Berger Nigeria	General Electric
Fuel arrangement	Utility arranges fuel	Project company provides fuel	Project company provides fuel	15-year fuel supply agreement with Seplat with a gas supply LC	Fuel supply agreement with Shell
Debt-equity Ratio				80/20	
Local shareholder equity (Entity, US\$ Million)	5%	0/100	0/100	Main equity sponsors: Azura-Edo Ltd 97.5% comprising APHL 50% (Amaya Capital 80%, American Capital 20%); AIM 30%; ARM 6%; Aldwych 14%; and Edo State 2.5%	
Foreign shareholder equity (Entity, US\$ Million)		20%	.45	Main equity sponsors: Azura-Edo Ltd 97.5% comprising APHL 50% (Amaya Capital 80%, American Capital 20%); AIM 30%; ARM 6%; Aldwych 14%; and Edo State 2.5%	
DFI Agency and financing method	The \$120 million financing was funded from a consortium of four commercial banks and three development finance institutions. The development financial institutions are FMO (Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V.), African Export-Import Bank, and DEG (Deutsche Investitions und Entwicklungsgesellschaft mbH). The commercial banks are Africa Merchant Bank (France), a division of Belgolaise Bank, United Bank for Africa (Nigeria), Rand Merchant Bank (South Africa), and Diamond Bank (Nigeria)			KfW Bankengruppe of Germany, the Netherlands Development Finance Company (FMO), International Finance Corporation (IFC), German Investment Corporation (DEG), French Investment Corporation, Emerging Africa Infrastructure Fund, the World Bank Group and Swedfund, OPIC	Subordinated debt: IFC, EIB, and Emerging Africa Infrastructure Fund and 4mm IFC equity
Total DFI financing (US\$ Million)	60.0	–	–	332.5	4.0
ODA Grants (US\$ Million)					
Local credit enhancements & security arrangements	Sovereign guarantee, US\$60 million Letter of Credit from Ministry of Finance	PPA backed by Nigerian Petroleum Development Company's oil revenues	PPA backed by Nigerian Petroleum Development Company's oil revenues		
Foreign credit enhancements & security arrangements	OPIC political risk insurance			Credit Enhancement PRG (IBRD)	

Table A11

IPP investments in Rwanda by project.

Project Information	Project Name
	KivuWatt
Capacity (MW)	100
Technology	Methane Gas
Total Investment (US\$ Million)	200.0
Year of financial closure	2011
COD	2015
Project status	Construction
Procurement method	DN
Number of bids	
Contract period	25
Contract type	BOO
Sponsors/Developer	ContourGlobal (100%/United States)
EPC	
Fuel arrangement	
Debt-equity Ratio	
Local shareholder equity (Entity, US\$ Million)	
Foreign shareholder equity (Entity, US\$ Million)	
DFI Agency and financing method	MIGA (Guarantee/\$26 Million/2011), AFDB (Loan/\$25 Million/2011) UK, Dutch, Swedish and Swiss governments loaned \$91mn
Total DFI financing (US\$ Million)	116.0
ODA Grants (US\$ Million)	–
Local credit enhancements & security arrangements	
Foreign credit enhancements & security arrangements	

Table A12
IPP investments in Senegal by project.

Project Information	Project Name	Project Name2	Project Name3	Project Name4	Project Name5
	GTI Dakar Ltd.	Kounoune I IPP	Saint-Louis - Dagana - Podor Rural Electrification	Sendou	Tobene
Capacity (MW)	52	67.5	19	125	87.5
Technology	OCGT+CCGT	MSD/HFO	Solar, PV	Coal	MSD/HFO
Total Investment (US\$ Million)	65.0	110.0	22.0	254.3	163.5
Year of financial closure	1997	2005	2010	2013	2014
COD	2000	2008		2017	2015
Project status	Operational	Operational	Operational	Construction	Construction
Procurement method	ICB	ICB	ICB	ICB	ICB, then DN
Number of bids			2		1
Contract period	15	15	25		
Contract type	BOOT	BOO	BOT		
Sponsors/Developer	IFC, Sondel (Greenwich Air Service, Inc).	Melec Powergen (part of Matelec grp of companies) (Lebanon), Mitsubishi (Japan)	Office National de l'Electricite (73%/Morocco), International Finance Corporation (17%)		
EPC	MEGS (Mediterranean Electric Generating Services) - a joint venture between Sondel and General Electric	MHI Equipment Europe, France, (Member of Mitsubishi Heavy Industries Group)			
Fuel arrangement		During the project negotiations, the structure of the FSA and PPA were changed to turn the PPA into a tolling agreement			
		70/30			
Debt-equity Ratio					
Local shareholder equity (Entity, US\$ Million)					
Foreign shareholder equity (Entity, US\$ Million)					
DFI Agency and financing method	IFC (Loan/\$13 Million/1997), IFC (Equity/\$2 Million/1997), IFC (Syndication/\$3 Million/1997), IFC (Equity/\$1 Million/1998), IFC (Syndication/\$12 Million/1998), IFC (Quasi-equity/\$7 Million/1998), IFC (Risk management/\$1 Million/2002)	IDA (Guarantee/\$7 Million/2005), IDA (Loan/\$10 Million/2005), IFC (Loan/\$21 Million/2005); MIR also listed Proparco, AfDB, BOAD and CBAO	IFC (Equity/\$1 Million/2010)	African Development Bank (AfDB)- Netherlands Development Finance Company (FMO)	IFC lead arranger, Euro tranche=€78.5 M-€28.5 M A loan by IFC, and €50 M B loan (€25 M by FMO and €25 M by Emerging Africa Infrastructure Fund/EaIF), and a local tranche for the CFA equivalent of €13.5 M by BOAD
Total DFI financing (US\$ Million)	39.0	53.7	1.0	108.0	135.1
ODA Grants (US\$ Million)					
Local credit enhancements & security arrangements	Government guarantee, escrow account	Government guarantee, a letter of credit from Senelec			
Foreign credit enhancements & security arrangements	Credit insurance through a guarantee program of SACE, the Italian export credit agency, and a partial interest subsidy through the Mediocredito Central Subsidy Department (MCSD)	A PRG, but never signed by government			IDA PRG

Table A13

IPP investments in Sierra Leone by project.

Project Information	Project Name
	Addax Biomass Plant
Capacity (MW)	15
Technology	Biomass
Total Investment (US\$ Million)	30.0
Year of financial closure	2011
COD	2013
Project status	Operational
Procurement method	DN
Number of bids	
Contract period	
Contract type	BOO
Sponsors/Developer	Addax & Oryx Group (100%/ United Kingdom)
EPC	
Fuel arrangement	
Debt-equity Ratio	61/39
Local shareholder equity (Entity, US\$ Million)	
Foreign shareholder equity (Entity, US\$ Million)	
DFI Agency and financing method	AFDB (Loan/\$30 Million/2011)
Total DFI financing (US\$ Million)	30.0
ODA Grants (US\$ Million)	
Local credit enhancements & security arrangements	
Foreign credit enhancements & security arrangements	

Table A14
IPP investments in Tanzania by project.

Project Information	Project Name		Project Name2		Project Name3		Project Name4	
	Independent Power Tanzania Ltd		Songas - Songo Songo Gas to Power Project		Mtwara Region Gas-to-Power Project		Symbion	
Capacity (MW)	100		189		18		120	
Technology	MSD/HFO		CCGT		CCGT/CCGT		OCGT/CCGT	
Total Investment (US\$ Million)	127.2		316.0		32.0		123.2	
Year of financial closure	1997		2001		2005		2006	
COD	2002		2004		2007		2006, 2007	
Project status	Operational		Operational		Operational		Operational	
Procurement method	DN		ICB		ICB		DN	
Number of bids	2		2					
Contract period	20		20					
Contract type	BOO		BOO					
Sponsors/Developer	VIP Engineering and Marketing Ltd (Tanzania), MechMar Energy Sdn Bhd		TransCanada sold majority shares to AES (USA) in 1999 and AES sold majority shares to Globeleq (UK) in 2003[1]. All preferred equity shares were converted into "Loan Notes" in June 2009. Only common shares remain		Artumas Group Inc. (87%/Canada) FMO 13%		Expiry Oct 2014 Emergency/short-term Built by Richmond, sold to Downans, then to Symbion	
EPC			Larsen & Toubro (L&T)					
Fuel arrangement	IPTL imports fuel, which is a pass through to the utility		Songo Songo gas provided to project company at a rate of US \$55/MMBtu for turbines I-V and at US\$2.17 MMBtu for turbine VI		Fuel provided by a consortium that includes the project sponsor (has a 25.4% stake in the gas concession), at a charge of US\$5.00 per MMBtu, which is passed through to utility		TANESCO purchases natural gas and fuel is a pass through	
Debt-equity Ratio	0/100		70/30		0/100		0/100	
Local shareholder equity (Entity, US\$ Million)	VIP (Tanzania, 30% in kind), (disputed), have sought to sell shares		4.83		100% financed with balance sheet of shareholders			
Foreign shareholder equity (Entity, US\$ Million)	Mechmar (Malaysia, 70%) have sought to sell shares		5.67		100% financed with balance sheet of shareholders		Equity financed	
DFI Agency and financing method					FMO, 13% equity shareholder			
Total DFI financing (US\$ Million)	–		249.0		4.2		–	
ODA Grants (US\$ Million)	–		100.3		–		–	
Local credit enhancements & security arrangements	Sovereign guarantee, liquidity facility equivalent to 4 months capacity charge (but not yet established)		Escrow account: for first 115 MW, with the government matching every US\$1 spent by the project company; liquidity facility equivalent to 4 months capacity charge for the first 3 years, declining to 2 months starting in year 4 through the remaining years of the contract		Tariff Equalization Fund provided, a fixed-value account designed to make up the difference between the national tariff and the cost-based tariff (which would otherwise be charged to the final consumer) under the project		No government guarantees	
Foreign credit enhancements & security arrangements								

Table A15

IPP investments in Togo by project.

Project Information	Project Name
	Centrale thermique de Lome
Capacity (MW)	100
Technology	Triple fuel
Total Investment (US\$ Million)	196.0
Year of financial closure	2008
COD	2010
Project status	Operational
Procurement method	DN
Number of bids	
Contract period	25
Contract type	BOT
Sponsors/Developer	ContourGlobal (80%/United States), International Finance Corporation (20%)
EPC	
Fuel arrangement	
Debt-equity Ratio	
Local shareholder equity (Entity, US\$ Million)	
Foreign shareholder equity (Entity, US\$ Million)	
DFI Agency and financing method	IFC (Equity/Loan) and OPIC
Total DFI financing (US\$ Million)	161.0
ODA Grants (US\$ Million)	
Local credit enhancements & security arrangements	Payment Guarantee
Foreign credit enhancements & security arrangements	

Table A16
A. IPP investments in Uganda by project.

Project Information	Project Name	Project Name2	Project Name3	Project Name4	Project Name5	Project Name6
	Kasese Cobalt, Mubuku III	Kilembe Mines (Mubuku I)	Kakira cogeneration plant	Bujagali Hydro Project	ECO Ishasha Mini Hydropower Plant	Tronder/Bugoye Hydro Electric Power Project (Mubuku II)
Capacity (MW)	9.9	5.4	32	250	6.5	13
Technology	Hydro, Small (< 20 MW)	Hydro, Small (< 20 MW)	Waste/Bagasse	Hydro	Hydro, Small (< 20 MW)	Hydro, Small (< 20 MW)
Total Investment (US\$ Million)	22.5	16.2	56.0	860.0	14.0	65.7
Year of financial closure	1999	1975	2003	2007	2008	2008
COD	Operational	not operational	2013	2012	2011	2009
Project status	Operational	not operational	Operational	Operational	Operational	Operational
Procurement method	DN		DN/REFIT(PPA3)	ICB	DN	DN
Number of bids				3		
Contract period	20		20	30	30	20
Contract type			BOO	BOT	BOT	BOT
Sponsors/Developer	Blue Earth Refineries Inc(100% Uganda)	Government of Uganda (51%)	Madhvani Group (100%/Uganda)	BEL Ltd (Sitho Global Power (58%/United States), Aga Khan Fund (31%/Switzerland))	Eco Power (100%/Sri Lanka)	Tronder Power Limited (100%/Norway)
EPC			In-house/consultant			
Fuel arrangement						
Debt-equity Ratio				78/22	70/30	53/32 (14% grant by Government of Norway)
Local shareholder equity (Entity, US\$ Million)			Non-recourse			Tronder
Foreign shareholder equity (Entity, US\$ Million)			Balance sheet			Norfund
DFI Agency and financing method			EADB	MIGA (Guarantee/\$115 Million/2007), IFC (Loan/\$130Million/2007), IDA (Guarantee/\$115 Million/2007), ADB (Loan/\$110 Million/2007), EIB (Loan/\$130 Million/2007)		EAIF/FMO/Norway govt/Norfund
Total DFI financing (US\$ Million)	-	-	15.0	370.0	-	48.2
ODA Grants (US\$ Million)			GET FIT			
Local credit enhancements & security arrangements				Government payment guarantee		14% grant by Government of Norway
Foreign credit enhancements & security arrangements						Government payment guarantee

Table A16

B: IPP investments in Uganda by project.

Project Information	Project Name7	Project Name8	Project Name9	Project Name10	Project Name11	Project Name12
	Mpanga Hydro Power Project	Namanve Power Plant	Kinyara Cogeneration Plant	Buseruka/Hydromax Hydropower Plant	Tororo Power Station	Tororo Power Station
Capacity (MW)	18	50	7.5	9	16	34
Technology	Hydro, Small (< 20 MW)	MSD/HFO	Waste/Bagasse	Hydro, Small (< 20 MW)	MSD/HFO	MSD/HFO
Total Investment (US\$ Million)	27.0	74.0	29.0	27.0	41.5	41.5
Year of financial closure	2008	2008	2009	2009	2009	2012
COD	2011		2011	2012	2010	
Project status	Operational	Operational	Operational	Operational	Operational	Operational
Procurement method	DN	ICB	DN	DN	DN	
Number of bids		3				
Contract period	20	6	20	30	9	
Contract type	BOT	BOT	BOO	BOT	BOO	
Sponsors/Developer	South Asia Energy Management Systems (SAEMS) (100%/United States)	Jacobsen Elektro (100%/Norway)	Kinyara Sugar Group (100%/Uganda)	Hydromax Limited (100%/Uganda)	Electro-Maxx (100%/Uganda)	
EPC						
Fuel arrangement						
Debt-equity Ratio				70/30	60/40	
Local shareholder equity (Entity, US\$ Million)						
Foreign shareholder equity (Entity, US\$ Million)						
DFI Agency and financing method	EAIF (\$14 million)/FMO			AFDB (Loan/\$9 Million/2009)		
Total DFI financing (US\$ Million)	20.0	–	–	9.0	–	–
ODA Grants (US\$ Million)						
Local credit enhancements & security arrangements	Payment Guarantee	Payment Guarantee		Variable government payments	Government payment guarantee	
Foreign credit enhancements & security arrangements						

Table A16

C: IPP investments in Uganda by project.

Project Information	Project Name13	Project Name14	Project Name15	Project Name16	Project Name17	Project Name18
	Kakaka Hydropower Project	Rwimi Hydropower Project	Lubilia Hydropower Project	Muvumbe Hydropower Project	Nengo Bridge Hydropower Project	SAIL Cogen
Capacity (MW)	5	5.4	5.4	6.5	6.9	6.9
Technology	Hydro, Small (< 20 MW)	Hydro, Small (< 20 MW)	Hydro, Small (< 20 MW)	Hydro, Small (< 20 MW)	Hydro, Small (< 20 MW)	Waste/Bagasse
Total Investment (US\$ Million)	18.0	18.0	18.0	14.0	27.0	22.0
Year of financial closure	2015	2015	2015	2015		
COD	2016	2017	2017	2017		
Project status	Financing in process	Under Construction	Under Construction	Under Construction	Cancelled	Cancelled
Procurement method	REFiT	REFiT	REFiT	REFiT	REFiT	REFiT
Number of bids						
Contract period	20	20	20	20	20	20
Contract type	BOT	BOT	BOT	BOT	BOT	BOO
Sponsors/Developer	Frontier (Danish Private Equity Fund)	Eco Power (100%/Sri Lanka)	Frontier (Danish Private Equity Fund)	Vidullanka (100%/Sri Lanka)	Jacobsen Elektro (100%/Norway)	Sugar Allied Industries (Ugandan)
EPC						
Fuel arrangement						
Debt-equity Ratio		70/30	65/35			
Local shareholder equity						

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Table A16 (continued)

Project Information	Project Name13	Project Name14	Project Name15	Project Name16	Project Name17	Project Name18
	Kakaka Hydropower Project	Rwimi Hydropower Project	Lubilia Hydropower Project	Muvumbe Hydropower Project	Nengo Bridge Hydropower Project	SAIL Cogen
(Entity, US\$ Million)						
Foreign shareholder equity (Entity, US\$ Million)						
DFI Agency and financing method	EaIF/FMO		EaIF/FMO		EADB	
Total DFI financing (US\$ Million)	–	–	–	–	–	–
ODA Grants (US\$ Million)		GET FiT	GET FiT	GET FiT	GET FiT	GET FiT
Local credit enhancements & security arrangements						
Foreign credit enhancements & security arrangements						

Table A16

D: IPP investments in Uganda by project.

Project Information	Project Name19	Project Name20	Project Name21	Project Name22
	SAEMS Nyamwamba SHPP	Siti I/II Hydropower Project	Tororo North/South	Soroti Solar
Capacity (MW)	9.2	21.5	10	10
Technology	Hydro, Small (< 20 MW)	Hydro Large	Solar PV	Solar PV
Total Investment (US\$ Million)	34.0	48.0	18.0	18.0
Year of financial closure	2015	2015	2015x	2015x
COD	2017	2017–18	2017	2016
Project status	Construction started in 2014	Under Construction	Under Construction	Operational
Procurement method	REFIT	REFIT	ICB	ICB
Number of bids				
Contract period	20	20		
Contract type	BOT	BOT		
Sponsors/Developer	South Asia Energy Management Systems (SAEMS) (100%/United States)	Frontier (Danish Private Equity Fund)	Simba/Building Energy	Access/TSK
EPC				
Fuel arrangement				
Debt-equity Ratio	73/27	70/30	75/25	75/25
Local shareholder equity (Entity, US\$ Million)				
Foreign shareholder equity (Entity, US\$ Million)				
DFI Agency and financing method	Other (Loan/\$24 Million/2012) Out of which EaIF is 6 million	EaIF/FMO (5.3)	FMO	FMO
Total DFI financing (US\$ Million)	6.0	5.3	–	–
ODA Grants (US\$ Million)	GET FiT	GET FiT		
Local credit enhancements & security arrangements				
Foreign credit enhancements & security arrangements				

Table A17

IPP investments in Zambia by project.

Project Information	Project Name	Project Name2
	Ndola Energy	TATA Itezhi-Tezhi HPP
Capacity (MW)	50	120
Technology	MSD/HFO	Hydro
Total Investment (US\$ Million)	72.0	230.0
Year of financial closure	2012	2014
COD	2013	2016
Project status	Operational	Construction
Procurement method	DN	DN
Number of bids		
Contract period		25
Contract type		BOT
Sponsors/Developer	Subsidiary of Concordia Energy (Group of Mauritius)	Tata Enterprises (50%/India), Zambia Electric Supply Corporation (ZESCO) (50%/Zambia)
EPC		Chinese EPC/ICB for EPC
Fuel arrangement		
Debt-equity Ratio		
Local shareholder equity (Entity, US\$ Million)		
Foreign shareholder equity (Entity, US\$ Million)		
DFI Agency and financing method		EIB (Equity/\$18 Million/2011), 2014: a \$142 million loan by DBSA, Proparco, ADB, and FMO
Total DFI financing (US\$ Million)	–	162.0
ODA Grants (US\$ Million)		
Local credit enhancements & security arrangements		
Foreign credit enhancements & security arrangements		

Table A18
IPP investments in South Africa by project.

Project	Capacity (MW)	Technology	Total Investment (US\$ Million)	Financial closure	COD	Project status	Procurement method	Sponsors/ Developer	DFI Agency and financing method	Total DFI financing (US\$ Million)	Local credit enhancements & security arrangements
Bethlehem Hydro	7	Hydro, Small (< 20 MW)	13.7	2005	2009, 2012	Operational	DN	NuPlanet (26%/ Netherlands)	Other (Loan/\$5 Million/2005)	5.0	
Darling Wind Farm	5	Wind, Onshore	9.9	2006	2008	Operational	DN	Darling Independent Power Producer Pty Ltd (26%/ South Africa)		–	
Sasol SlimSun Swartland Solar Park	373	OCGT/CCGT Solar, PV	399.0	2010	2010	Operational	DN	Sasol	IDC (in 2012 ZAR) (Exchange rate .12)	8.9	Payment Guarantee
RustMol Solar Farm	6.9	Solar, PV	28.0	2012	2013	Operational	ICB		IDC (in 2012 ZAR) (Exchange rate .12)	9.8	Payment Guarantee
Konkoonistes Solar Energy Facility	9.7	Solar, PV	43.9	2012	2013	Operational	ICB		IDC (in 2012 ZAR) (Exchange rate .12)	15.0	Payment Guarantee
Aries Solar Energy Facility	9.7	Solar, PV	44.5	2012	2013	Operational	ICB		IDC (in 2012 ZAR) (Exchange rate .12)	15.0	Payment Guarantee
Greefspan PV Power Plant	9.9	Solar, PV	53.5	2012	2014	Operational	ICB		IDC (in 2012 ZAR) (Exchange rate .12)	10.0	Payment Guarantee
Mulilo Solar PV De Aar	10	Solar, PV	39.3	2012	2013	Operational	ICB		IDC (in 2012 ZAR) (Exchange rate .12)	13.6	Payment Guarantee
Herbert PV Power Plant	20.0	Solar, PV	105.3	2012	2013	Operational	ICB		IDC (in 2012 ZAR) (Exchange rate .12)	12.8	Payment Guarantee
Mulilo Solar PV	20	Solar, PV	79.1	2012	2015	Operational	ICB		IDC (in 2012 ZAR) (Exchange rate .12)	26.9	Payment Guarantee
Prieska Dasielklip Wind Energy Facility	27	Wind, Onshore	83.1	2012	2014	Operational	ICB		IDC (in 2012 ZAR) (Exchange rate .12)	18.0	Payment Guarantee
MetroWind Van Stadens Wind Farm	27	Wind, Onshore	74.8	2012	2014	Operational	ICB			–	Payment Guarantee
Soutpan Solar Park	28	Solar, PV	155.7	2012	2014	Operational	ICB			–	Payment Guarantee
Witkop Solar Park	30	Solar, PV	174.3	2012	2014	Operational	ICB			–	Payment Guarantee
Touwsrivier Solar Park	36	Solar, PV	197.5	2012	2014	Operational	ICB			–	Payment Guarantee
De Aar Solar PV	45.6	Solar, PV	178.0	2012	2014	Operational	ICB	Globeleq	DBSA in 2012 ZAR (Exchange rate .12)	43.0	Payment Guarantee
South Africa Mainstream Renewable Power	45.6	Solar, PV	173.6	2012	2014	Operational	ICB	Globeleq	DBSA in 2012 ZAR (Exchange rate .12)	41.9	Payment Guarantee
Droogfontein Khi Solar One	50	Solar CS	509.8	2012	2016	Operational	ICB		IPC,EIB, DBSA and IDC all have debt while IDC has 29% equity also	298.7	Payment Guarantee
Letsatsi Solar Photovoltaic Park	64	Solar, PV	320.9	2012	2014	Operational	ICB			–	Payment Guarantee
Lesedi Solar Photovoltaic Park	64	Solar, PV	322.7	2012	2014	Operational	ICB			–	Payment Guarantee
Hopefield Wind Farm	65.4	Wind, Onshore	195.6	2012	2014	Operational	ICB			–	Payment Guarantee

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Table A18 (continued)

Project	Capacity (MW)	Technology	Total Investment (US\$ Million)	Financial closure	COD	Project status	Procurement method	Sponsors/ Developer	DFI Agency and financing method	Total DFI financing (US\$ Million)	Local credit enhancements & security arrangements
Kalkbult	72.5	Solar, PV	274.9	2012	2013	Operational	ICB		DBSA in 2012 ZAR (Exchange rate .12)	29.8	Payment Guarantee
Kathu Solar Plant	75	Solar, PV	430.4	2012	2014	Operational	ICB		DBSA in 2012 ZAR (Exchange rate .12)	45.0	Payment Guarantee
Solar Capital De Aar	75	Solar, PV	296.6	2012	2014	Operational	ICB			–	Payment Guarantee
Nobelsfontein Phase 1	75	Wind, Onshore	196.8	2012	2014	Operational	ICB			–	Payment Guarantee
Kouga Wind Farm	80	Wind, Onshore	235.6	2012	2014	Operational	ICB		IDC (in 2012 ZAR) (Exchange rate .12)	53.9	Payment Guarantee
Dorper Wind Farm	97.5	Wind, Onshore	286.1	2012	2014	Operational	ICB			–	Payment Guarantee
KaXu Solar One	100	Solar CS	976.3	2012	2014	Operational	ICB		DBSA 1,171,290,600, IDC 829,656,566, IFC 600,000,000, IFC (as Implementation Entity of the Clean Technology Fund)	454.8	Payment Guarantee
Jeffreys Bay	138	Wind, Onshore	366.5	2012	2014	Operational	ICB		232,405,000, MEZZ Debt: DBSA 195,312,707, IDC 195,312,707 Equity: IDC 29%	101.8	Payment Guarantee
Cookhouse Wind Farm	138.6	Wind, Onshore	295.6	2012	2014	Operational	ICB	Globeleq	DBSA 848,700,000	–	Payment Guarantee
Vredendal Solar Park	8.82	Solar, PV	29.1	2013		Operational	ICB			–	Payment Guarantee
Stortemelk Hydro (Pty) Ltd	4.4	Hydro Small (< 20 MW)	17.4	2013		Operational				–	Payment Guarantee
Upington Solar PV	8.9	Solar, PV	26.5	2013		Operational	ICB			–	Payment Guarantee
Aurora-Rietvlei Solar Power	9	Solar, PV	30.3	2013		Operational	ICB			–	Payment Guarantee
Neusberg Hydro	10	Hydro, Small (< 20 MW)	73.5	2013		Operational	ICB		IDC (in 2013 ZAR) (Exchange rate .12) senior and mezz debt	19.7	Payment Guarantee
Electric Project A	21	Wind, Onshore	54.4	2013		Operational	ICB		IDC in 2013 ZAR (Exchange rate .12)	15.5	Payment Guarantee
Chaba Wind Farm	21	Wind, Onshore	69.7	2013	2016	Operational	ICB		IDC in 2013 ZAR (Exchange rate .12)	19.9	Payment Guarantee
Waainek Wind Power	23.3	Wind, Onshore	147.2	2013		Operational	ICB		IDC 25% equity	–	Payment Guarantee
Linde	36.8	Solar, PV	642.2	2013		Construction	ICB			45.1	Payment Guarantee
Bokpoort CSP Project	50	Solar CS	161.3	2013		Operational	ICB		IDC 2013	46.1	Payment Guarantee
Grassridge Wind Energy Project	59.8	Wind, Onshore	312.0	2013		Operational	ICB		OPIC	222.7	Payment Guarantee
Boshof Solar Park	60	Solar, PV	286.6	2013		Operational	ICB			–	Payment Guarantee
Dreunberg	69.6	Solar, PV	294.8	2013	2014	Operational	ICB		IDC	111.1	Payment Guarantee
Sishen Solar Facility	74	Solar, PV	326.9	2013		Operational	ICB			–	Payment Guarantee
Solar Capital De Aar 3	75	Solar, PV	290.7	2013		Operational	ICB		DBSA	60.0	Payment Guarantee
Jasper Power Company	75	Solar, PV	252.1	2013		Operational	ICB		DBSA	44.1	Payment Guarantee
West Coast One	90.8	Wind, Onshore	365.9	2013	2016	Operational	ICB			–	Payment Guarantee
Wind Farm	94.8	Wind, Onshore		2013		Operational	ICB				
Tsitsikamma Community		Wind, Onshore									
Wind Farm		Wind, Onshore									
Anakhala Emoyeni	133.7	Wind, Onshore	497.0	2013	2016	Operational	ICB	IFC		76.1	Payment Guarantee

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Table A18 (continued)

Project	Capacity (MW)	Technology	Total Investment (US\$ Million)	Financial closure	COD	Project status	Procurement method	Sponsors/ Developer	DFI Agency and financing method	Total DFI financing (US\$ Million)	Local credit enhancements & security arrangements
Wind Farm											
Gouda Wind Project Mkuze	135.5 16.5	Wind, Onshore Biomass	336.3 95.6	2013 2015		Operational	ICB			–	Payment Guarantee
Johannesburg Landfill Gas to Electricity	18	Landfill Gas	24.8	2014		Financial and Approvals underway Partially Operational	ICB			–	Payment Guarantee
Tom Burke Solar Park	60	Photovoltaic Thin Film Fixed		2014	2016	Operational	ICB			–	Payment Guarantee
Adams Solar PV 2	75	Photovoltaic Crystalline Fixed		2014		Construction	ICB			–	Payment Guarantee
Electra Capital (Pty) Ltd	75	Photovoltaic Crystalline Fixed		2014	2016	Operational	ICB			–	Payment Guarantee
Mulilo Sonnedix	75	Photovoltaic Crystalline Fixed	108.0	2014	2016	Operational	ICB			–	Payment Guarantee
Prieska PV	75	Photovoltaic Crystalline Fixed	200.0	2014		Operational	ICB			–	Payment Guarantee
Mulilo Prieska PV	75	Photovoltaic Crystalline- Single Axis		2014		Operational	ICB		IDC in 2014	20.2	Payment Guarantee
Pulido Solar Park	75	Photovoltaic Thin Film Fixed		2014		Financing Done	ICB			–	Payment Guarantee
Noupoort Mainstream Wind	80	Wind, Onshore	180.0	2014	2016	Operational	ICB		EKF and DBSA	108.5	Payment Guarantee
Nojoli Wind Farm	86.6	Wind, Onshore		2014	2017	Operational	ICB			–	Payment Guarantee
Longyuan Mulilo De Aar	96.5	Wind, Onshore	180.0	2014		Construction	ICB		IDC	63.0	Payment Guarantee
Maanhaarberg Wind Energy Facility											
Ilanga CSP 1/ Karoshoek Solar One	100	Concentrated Solar Power, parabolic trough, with storage (4.5 h per day)	735.4	2014		Construction	ICB		IDC and DBSA	180.0	Payment Guarantee
Xina Solar One	100	Concentrated Solar Power, parabolic trough, with storage (5 h per day)	880.0	2014		Construction	ICB		DBSA 800,000,000, IDC 750,000,000, AfDB 1,500,000,000 and IDC 20% equity	316.8	Payment Guarantee
Red Cap - Gibson Bay	110	Wind, Onshore	202.5	2014		Construction	ICB			–	Payment Guarantee
Khobab Wind Farm	137.7	Wind, Onshore	315.0	2014		Construction	ICB		DBSA, EKF	214.2	Payment Guarantee
Loeriesfontein 2	138.2	Wind, Onshore	315.0	2014		Construction	ICB		DBSA, EKF	208.5	Payment Guarantee
Wind Farm											
Longyuan Mulilo De Aar 2 North	139.0	Wind, Onshore	264.6	2014		Construction	ICB		IDC	85.5	Payment Guarantee

Notes:

- a) REIPPPP investment data is derived from public sources and has an error range of around 10% and that final financial close data is different from bid data and is not yet publicly available
b) Cells with no input represent information we do not have. Investment number cells with a "-" sign mean 0 investment.

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